Background-Annihilated Target-Constrained Interference-Minimized Filter (TCIMF) for Hyperspectral Target Detection

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***Abstract*— The target-constrained interference-minimized filter (TCIMF) has been widely used in various target detection applications for hyperspectral data exploitation. However, like other classic target detection algorithms, the complex background (BKG) of a scene significantly impacts its performance. To better cope with BKG, this article develops a BKG-annihilated TCIMF (BA-TCIMF) that can be implemented in two stages with BKG annihilation in the first stage followed by target detectability (TD) enhancement and target BKG suppression performed by TCIMF in the second stage. In particular, the second stage extracts additional BKG signatures from the BA data as unwanted signatures to enhance TD via orthogonal subspace projection (OSP) while suppressing target BKG in the BA data by constrained energy minimization (CEM). Depending upon how these two stages are carried out, three versions of BA-TCIMF, data sphered BA-TCIMF (DS-BA-TCIMF), low-rank and sparse matrix decomposition (LRaSMD) BA-TCIMF (LRaSMD-BA-TCIMF), and component decomposition analysis-BA-TCIMF (CDA-BA-TCIMF), are derived. Experimental results demonstrate that BA-TCIMF performs as it is designed and better than many existing target detection algorithms.**

***Index Terms*— Background (BKG), BKG-annihilated (BA), constrained energy minimization (CEM), go decomposition (GoDec), low-rank and sparse matrix decomposition (LRaSMD), target-constrained interference-minimized filter (TCIMF).**

NOMENCLATURE

2OS	Second-order statistics.
AUC	Area under an ROC curve.
BA	BKG-annihilated.
BKG	Background.
BS	BKG suppression.
CEM	Constrained energy minimization.
CDA	Component decomposition analysis.
CRD	Collaborative representation detector.
DS	Data sphering.
ECEM	Ensemble cascaded CEM.
GoDec	Go decomposition.
HCEM	Hierarchical CEM.
IC	Independent component.
KTCIMF	Kernel TCIMF.
LRaSMD	Low-rank and sparse matrix decomposition.
LRR	Low-rank representation.
LSDM	Low-rank and sparse decomposition model.
MX-SVD	Minimax singular value decomposition.
ODP	Overall detection probability.
OSP	Orthogonal subspace projection.
PC	Principal component.
ROC	Receiver operating characteristic.
SNPR	Signal-to-noise probability ratio.
TBS	Target background suppression.
TDE	Target detectability enhancement.
TD	Target detectability.
TCIMF	Target-constrained interference-minimized filter.
VD	Virtual dimensionality.

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I. INTRODUCTION

HYPERSPECTRAL imagery is a data cube, which is acquired by three dimensions, two spatial dimensions, and one spectral dimension. Thus, every pixel is indeed a vector of which each component corresponds to the measurement of a spectral value acquired by a specific spectral wavelength [1], [2], [3], [4]. The very fine spectral resolution provided by hundreds of contiguous spectral bands allows a hyperspectral imager to uncover and reveal many subtle material substances, which cannot be identified by prior knowledge or visual inspection in many applications, specifically, subpixel detection, anomaly detection, endmember finding, mixed pixel classification and quantification, and so on.

Over the last decades, a variety of target detection methods have been developed. For example, Chang [1], [6] and Harsanyi and Chang [5] developed a target detection approach from a mixed-signal detection that annihilated undesired signals by OSP before target detection takes place. Technically speaking, the key success for OSP is its use of an orthogonal subspace projector, $P_{\mathbf{U}}^{\perp}$, which annihilates the signal sources specified by an undesired target signature matrix $\mathbf{U}$. However, it is very sensitive to the knowledge of $\mathbf{U}$. Since many unknown signals cannot be known *a priori* to be included in $\mathbf{U}$, this limits the utilization of OSP. To mitigate this dilemma, Harsanyi first developed a CEM approach in his dissertation [7] and latter in an application [8]. It constrained

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the target to be detected while suppressing all other signal sources by inverting the sample correlation matrix to enhance target detection with more details provided in [9]. By taking advantage of OSP and CEM, Ren and Chang [10] combined OSP and CEM to derive TCIMF, which used OSP to annihilate undesired targets to increase TD and in the meantime used CEM to improve TBS simultaneously. All these three algorithms are included in the popular software Environment for Visualizing Images (ENVI) that is developed by L3Harris Geospatial and has been widely used for remote sensing image analysis [11].

One major strength of TCIMF is its ability in remedying the deficiency of OSP by including the strength of CEM, which suppresses unknown BKG via inverting the sample correlation matrix. Nevertheless, TCIMF still has two inherent issues arising from the undesired signature matrix $\mathbf{U}$, which must be known *a priori* like OSP. In many real-world applications, this assumption is not practical due to the fact that hyperspectral imaging can uncover many unknown signal sources, which cannot be identified by prior knowledge, specifically, BKG. A second issue is that TCIMF is also very sensitive to the knowledge of $\mathbf{U}$. If $\mathbf{U}$ is provided precisely, TCIMF works very effectively. On the other hand, if $\mathbf{U}$ is not reliable, then TCIMF may not perform, and it is designed. To further alleviate this problem, TCIMF was subsequently extended by a signal decomposition interference-annihilated (SDIA) approach [12], which used a signal decomposition interference noise (SDIN) model to generate a set of unknown interfering signature sources specified by Ψ that can be found by automatic target generation process (ATGP) [13]. Using Ψ as additional interfering signal sources in conjunction with $\mathbf{U}$, TCIMF improves its detection performance for better interference annihilation. Unfortunately, with significantly improved spatial and spectral resolutions, a hyperspectral image scene becomes more complicated than a multispectral image scene. The ATGP-generated interfering signal sources Ψ used by the SDIN model may not be capable of taking care of the BKG issue effectively if Ψ is unable to faithfully represent BKG signatures.

In order to deal with BKG more effectively, we look closely into the structure of TCIMF that is actually designed to tackle two types of BKG. One is the global BKG that can be suppressed by inverting $\mathbf{R}$ via CEM. The other is target BKG that can be suppressed by annihilating $\mathbf{U}$ via OSP. The SDIA proposed in [12] extended TCIMF by introducing interfering signal sources Ψ as unwanted signal sources to be annihilated by OSP, so as to enhance target detection. However, both TCIMF and SDIA did not look into the issue of how to effectively suppress the global BKG. This article investigates these two types of BKG and generalizes TCIMF from a rather different perspective.

To annihilate the global BKG, three approaches are of particular interest. One is to assume that BKG can be characterized by the first two orders of data statistics (2OS). In this case, DS [2, Ch. 6] is then applied to annihilate BKG prior to target detection. This DS was further used to expand OSP's capability in improving target detection [14] and anomaly detection [15]. A second one is to represent a data space $\mathbf{X}$ as a decomposition of a low-rank subspace $\mathbf{L}$ and sparse subspace $\mathbf{S}$ so that $\mathbf{L}$ and $\mathbf{S}$ can be used to specify the BKG and target spaces, respectively. One of the most studied techniques in recent years is the LRaSMD proposed in [16], which decomposes a data space $\mathbf{X}$ as a three-component space, low-rank space $\mathbf{L}$, sparse space $\mathbf{S}$, and noise space $\mathbf{N}$. Unlike LRaSMD, a third approach was recently developed by Chen *et al.* [17], called CDA that decomposed a data space into PC, IC, and noise components.

In this article, we integrate the above three individual approaches into TCIMF to derive BA-TCIMF to yield the global BA data space, $\mathbf{X}_{\mathrm{BA}}$. We then use an unsupervised method to generate unwanted BKG signatures from $\mathbf{X}_{\mathrm{BA}}$ for TBS so as to improve TD. In essence, BA-TCIMF makes use of two preprocessing steps to deal with two types of BKG issues: global BKG annihilation and target BKG annihilation prior to TCIMF for target detection.

In order to further explore the nonlinear structures of TCIMF a kernel version of TCIMF, KTCIMF was generalized in [18] and can be used to extend BA-TCIMF to derive BA-KTCIMF. Unfortunately, due to the fact that TCIMF already significantly improves its detection performance, the advantages offered by KTCIMF will be limited. Accordingly, the benefits gained by KTCIMF are not well-paid off at the expense of its exceedingly high computational complexity, which will be shown in the experiments conducted in this article. Nevertheless, BA-KTCIMF can still manage to improve KTCIMF.

It has been shown in [19] that the commonly used 2-D ROC curve of detection probability P_D versus false alarm probability P_F was ineffective because both P_D and P_F are determined by the same threshold and cannot stand alone to be used as independent criteria. Specifically, when P_D is high, P_F is also high and vice versa. Consequently, using the AUC of the 2-D ROC curve of (P_D, P_F) cannot be used to evaluate BS. In particular, Chang [19] introduced a 3-D ROC curve from which three 2-D ROC curves can be derived, i.e., the 2-D ROC curve of (P_D, P_F), the 2-D ROC curve of (P_D, τ), and the 2-D ROC curve of (P_F, τ). Based on these three 3-D ROC curve-derived 2-D ROC curves, eight detection measures were designed to evaluate AD performance in terms of TD, BS, joint TD and BS (TDBS), and overall detection performance.

Several novelties are summarized in the following.

1) It introduces a new concept of BA into TCIMF to derive BA-TCIMF, which can accomplish global BKG annihilation and target BKG annihilation.
2) It uses DS, LRaSMD, and CDA to accomplish global BKG annihilation.
3) It further implements an unsupervised means to find unwanted BKG signatures for TBS.
4) It allows BA-TCIMF to detect targets and suppress BKG in different spaces.
5) It does not need training samples to achieve BA as many deep learning (DL) or model-based works reported in the literature.

The above novelties yield the following main contributions.

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1) It generalizes TCIMF to BA-TCIMF in two-stage processes with the first stage for global BKG annihilation followed by the second stage to perform TDE and TBS.
2) It derives three versions of BA-TCIMF, DS-BA-TCIMF, LRaSMD-BA-TCIMF, and CDA-BA-TCIMF.
3) It also uses MX-SVD to extract BKG signatures from the low-rank space $\mathbf{L}$ for DS-BA-TCIMF, LRaSMD-BA-TCIMF, and CDA-BA-TCIMF as additional unknown BKG signatures, so as to achieve better TDE and TBS.
4) Finally, it extends BA-TCIMF to training sampling-based BA-TCIMF to study the robustness of TCIMF to its training sample-calculated target signature matrix $\mathbf{D}$.

The remainder of this article is organized as follows. Section II briefly reviews related works. Section III derives BA-TCIMF that has three versions of BA-TCIMF, DS-BA-TCIMF, LRaSMD-BA-TCIMF, and CDA-BA-TCIMF. Section IV describes the datasets to be studied along with 3-D ROC curve-derived detection measures to address the BKG issue in target detection. Section V performs extensive experiments and discusses their results. Section VI extends BA-TCIMF to training sampling-based BA-TCIMF and BA-KTCIMF. Section VII summarizes conclusions.

II. Related Works

Target detection has been studied extensively and comprehensively in [20] where various target detectors have been derived. Among them, CEM and OSP are two of most promising target detectors where many variants were also derived, such as SDIA in [12], ATGP in [13], OSP-TD in [14], linearly constrained minimum variance (LCMV) in [9], TCIMF [10], HCEM [21], ECEM [22], CEM anomaly detector (CEMAD) [23], and kernel CEM (KCEM) [24]. Interestingly, TCIMF is the only one that combines the strengths of both OSP and CEM, and is the main base detector to derive BA-TCIMF in this article.

A. Target Constrained Interference Minimize Filter

Comparing to OSP [6], the main strength of CEM [7] is to pass the desired signal while minimizing output energy of unknown signal sources without obtaining them. However, TCIMF [10] needs the undesired signal as a prior, which limited its application in the practices because the undesired signals usually include unknown signal sources that are almost impossible to be acquired. In target detection applications, signal sources can be divided into two categories: one of them is targets of interest as the desired targets, and the targets of no interest as unwanted targets, such as undesired signal sources by prior knowledge. Naturally, eliminating or suppressing these unwanted signal sources will lead to an increase in the detectability of targets of interest. To achieve this goal, the idea of the TCIMF can be described as the following procedure. Let the known signals be decomposed into interesting signals, $\mathbf{D}$, denoted by $\mathbf{D} = [\mathbf{d}_1\mathbf{d}_2, \ldots, \mathbf{d}_p]$ and uninteresting signals, $\mathbf{U}$, denoted by $\mathbf{U} = [\mathbf{u}_1\mathbf{u}_2, \ldots, \mathbf{u}_q]$ as an undesired target signature matrix. Now, if we let

$$\mathbf{c} = \begin{bmatrix} \mathbf{1}_{p\times 1} \\ \mathbf{0}_{q\times 1} \end{bmatrix}$$

constrain these two target matrices by

$$(\mathbf{DU})^T\mathbf{w} = \begin{bmatrix} \mathbf{1}_{p\times 1} \\ \mathbf{0}_{q\times 1} \end{bmatrix} = \left(\underbrace{1,\ldots,1}_{p},\underbrace{0,\ldots,0}_{q}\right)^T \tag{1}$$

where $\mathbf{DU} = \left[\mathbf{d}_1\mathbf{d}_2, \ldots, \mathbf{d}_p\mathbf{u}_1\mathbf{u}_2, \ldots, \mathbf{u}_q\right]$, $\mathbf{1}_{p\times 1}$ is a $p \times 1$ vector with ones in all components, and $\mathbf{0}_{q\times 1}$ is a $q \times 1$ vector with zero in its components, where 1 represents the desired targets and 0 represents the undesired targets. Using (14) as a constraint vector, TCIMF can be formulated as the following multidesired target-constrained and multiundesired target-annihilated optimization problem:

$$\min_{\mathbf{w}}\left\{\mathbf{w}^T\mathbf{R}\mathbf{w}\right\} \quad \text{s.t.} \quad (\mathbf{DU})^T\mathbf{w} = \begin{bmatrix} \mathbf{1}_{p\times 1} \\ \mathbf{0}_{q\times 1} \end{bmatrix} \tag{2}$$

where $\mathbf{R}$ is the global correlation matrix, and the optimal weight vector $\mathbf{w}*$ can be obtained by

$$\mathbf{w}^* = \mathbf{R}^{-1}\mathbf{DU}\left((\mathbf{DU})^T\mathbf{R}^{-1}\mathbf{DU}\right)^{-1}\begin{bmatrix} \mathbf{1}_{p\times 1} \\ \mathbf{0}_{q\times 1} \end{bmatrix}. \tag{3}$$

The FIR filter coefficient specified by w^* given above is called the TCIMF given by

$$\delta^{\text{TCIMF}}(\mathbf{r}) = \left(\mathbf{R}^{-1}\mathbf{DU}\left((\mathbf{DU})^T\mathbf{R}^{-1}\mathbf{DU}\right)^{-1}\begin{bmatrix} \mathbf{1}_{p\times 1} \\ \mathbf{0}_{q\times 1} \end{bmatrix}\right)^T\mathbf{r} \tag{4}$$

with $\mathbf{w}^{\text{TCIMF}} = \mathbf{w}^*$ specified by (3). Now, if we assume that $\mathbf{d}$ is the desired target signature and let $\mathbf{D} = \mathbf{d}$ and $\mathbf{U} = \emptyset$, then (2) is reduced to

$$\min_{\mathbf{w}}\left\{\mathbf{w}^T\mathbf{R}\mathbf{w}\right\} \quad \text{s.t.} \quad \mathbf{d}^T\mathbf{w} = 1 \tag{5}$$

which yields the CEM subpixel target detector derived in [7]

$$\delta^{\text{CEM}}(\mathbf{r}) = \frac{\mathbf{d}^T\mathbf{R}^{-1}\mathbf{r}}{(\mathbf{d}^T\mathbf{R}^{-1}\mathbf{d})}. \tag{6}$$

On the other hand, if we assume that $\mathbf{d}$ is the desired target signature and let $\mathbf{D} = \mathbf{d}$, $\mathbf{U} = \emptyset$, and $\mathbf{R}^{-1} = P_{\mathbf{U}}^{\perp}$, then (2) is reduced to OSP-TD

$$\delta^{\text{OSP}}(\mathbf{r}) = \frac{\mathbf{d}^T P_{\mathbf{U}}^{\perp}\mathbf{r}}{\left(\mathbf{d}^T P_{\mathbf{U}}^{\perp}\mathbf{d}\right)} \tag{7}$$

where $P_{\mathbf{U}}^{\perp}$ is the OSP annihilation projection matrix given by

$$P_{\mathbf{U}}^{\perp} = \mathbf{I} - \mathbf{U}\left(\mathbf{U}^T\mathbf{U}\right)^{-1}\mathbf{U}^T. \tag{8}$$

It is equivalent to replacing $\mathbf{R}^{-1}$ in (6) with $P_{\mathbf{U}}^{\perp}$.

B. Matrix Decomposition for Global BKG Annihilation

To annihilate the global BKG, three approaches are of particular interest.

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1) DS: It assumes that global BKG can be characterized by the first two orders of data statistics (2OS), mean μ, and covariance matrix $\mathbf{K}$. The DS is performed by mapping the original data sample vector $\mathbf{r}$ to its sphered data sample vector $\hat{\mathbf{r}}$ in the sphered data space $\hat{\mathbf{X}}$ by

$$\hat{\mathbf{r}} = \mathbf{K}^{-1/2}(\mathbf{r} - \mu) \tag{9}$$

where $\mathbf{K}^{-1/2}$ is referred to as whitening matrix in [26]. Details of DS can be found in [2, Sec. 6.3.1] along with its code [2, pp. 1000 and 1001]. It is interesting to see that the well-known anomaly detector developed by Reed and Yu [25], RXD, is given by

$$\delta^{\mathrm{RXD}}(\mathbf{r}) = (\mathbf{r} - \mu)^T \mathbf{K}^{-1}(\mathbf{r} - \mu) \tag{10}$$

which happens to be the Mahalanobis distance between $\mathbf{r}$ and μ. Now, if we rewrite $\mathbf{K}^{-1}$ as $\mathbf{K}^{-1} = \mathbf{K}^{-1/2}\mathbf{K}^{-1/2}$, then, using (9), (10) can be reexpressed

$$\left(\mathbf{K}^{-1/2}(\mathbf{r} - \mu)\right)^T \mathbf{K}^{-1/2}(\mathbf{r} - \mu) = ||\mathbf{K}^{-1/2}(\mathbf{r} - \mu)||^2 = ||\hat{\mathbf{r}}||^2 \tag{11}$$

which turns out to be the vector length of $\hat{\mathbf{r}}$. This shows that, for each data sample vector $\mathbf{r}$, RXD is nothing more than finding the vector length of its sphered data sample vector $\hat{\mathbf{r}}$.

2) Matrix Decomposition: Another is to represent a data space $\mathbf{X}$ as a decomposition of a low-rank subspace $\mathbf{L}$ and sparse subspace $\mathbf{S}$ so that $\mathbf{L}$ and $\mathbf{S}$ can be used to specify the BKG and target spaces, respectively. This approach has been one of the most studied techniques in recent years.

a) RPCA: One earlier approach was robust PC analysis (RPCA) developed by Candès *et al*. [27], which decomposed the data matrix $\mathbf{X}$ into a matrix decomposition

$$\mathbf{X} = \mathbf{L} + \mathbf{S} \tag{12}$$

where $\mathbf{L}$ is a low-rank matrix and $\mathbf{S}$ has an arbitrary magnitude where the given data $\mathbf{X}$ are assumed to be of low inherent dimensionality. RPCA can cope with the noise of extremely high amplitude by PC pursuit (PCP) estimate that solves

$$\min \|\mathbf{L}\|_* + \lambda \|\mathbf{S}\|_{l1} \quad \text{s.t.} \quad \mathbf{L} + \mathbf{S} = \mathbf{X} \tag{13}$$

where $\|\mathbf{L}\|_*$ is defined by the nuclear norm.

Interestingly, Vaswani *et al*. developed an approach in [28] and [29], called robust subspace learning (RSL), which treated an outlier as an additive sparse corruption via sparse + low-rank (S+LR) matrix decomposition. Using this interpretation, RPCA can be considered as a variant of RSL via S + LR formulation.

b) LRaSMD: One major problem with (12) is that noise and sparse signals are blended in $\mathbf{S}$. As a result, on one end, noise may be mistaken for targets that result in false alarms. On another end, targets may be treated as noise, which causes missing targets. Either case will degrade detection performance. To resolve this issue, a more attractive model, called the LRaSMD, was proposed in [16], which decomposes a data space $\mathbf{X}$ as a three-component space

$$\mathbf{X} = \mathbf{L} + \mathbf{S} + \mathbf{N} \tag{14}$$

where an algorithm, called GoDec, was designed to produce the low-rank space $\mathbf{L}$, sparse space $\mathbf{S}$, and noise space $\mathbf{N}$. Recently, GoDec was further extended to OSP-GoDec in [30]. As a result of (14), OSP can be operated on the low-rank space $\mathbf{L}$ via an OSP annihilator $P_{\mathbf{L}}^{\perp}$, which can be obtained from (8) by replacing $\mathbf{U}$ with $\mathbf{L}$ as

$$\mathbf{X}_{\mathbf{L}}^{\perp} = P_{\mathbf{L}}^{\perp}\mathbf{X} \tag{15}$$

for BKG annihilation onto the space $\mathbf{X}_{\mathbf{L}}^{\perp}$ orthogonal to the low-rank space $\mathbf{L}$ to annihilate $\mathbf{L}$, which is considered as the BKG space.

c) CDA: A third approach is recently developed by Chen *et al.* [17], called CDA, which represents a data space as a decomposition of PC, IC, and noise component spaces by

$$\mathbf{X} = \mathbf{PC} + \mathbf{IC} + \mathbf{N}. \tag{16}$$

By assuming that BKG is characterized by the PC space, $\mathbf{PC}$, BKG annihilation can be accomplished by

$$\mathbf{X}_{\mathbf{PC}}^{\perp} = P_{\mathbf{PC}}^{\perp}\mathbf{X} = (\mathbf{I} - P_{\mathbf{PC}})\mathbf{X} \tag{17}$$

with $P_{\mathbf{PC}} = \mathbf{PC}((\mathbf{PC})^T\mathbf{PC})^{-1}(\mathbf{PC})^T$.

C. GoDec

Inspired by matrix decomposition and RPCA, GoDec develops its theory to manipulate the low-rank approximation and sparse presentation. It first projects an input dense matrix on its columns space, $\mathbf{Y}_1 = \mathbf{X}\mathbf{A}_1$, and then, its rows space, $\mathbf{Y}_2 = \mathbf{X}^T\mathbf{A}_2$, respectively, where $\mathbf{A}_1$ and $\mathbf{A}_2$ are random matrices. The projection process is a so-called bilateral random projection (BRP). This process is similar to Halko's randomized approximate matrix decomposition [31] except for the number of projections. Thus, the low-rank matrix is given by

$$\mathbf{L} = \mathbf{Y}_1(\mathbf{A}_2\mathbf{Y}_1)^{-1}\mathbf{Y}_2. \tag{18}$$

After obtaining the low-rank approximation, GoDec uses the model

$$\mathbf{X} = \mathbf{L} + \mathbf{S} + \mathbf{N} \quad \text{s.t.} \quad \text{rank}(\mathbf{L}) \le r, \quad \text{card}(\mathbf{S}) \le k \tag{19}$$

where card($\mathbf{S}$) is defined as sparsity cardinality (SC) of $\mathbf{S}$. Recall that RPCA also uses the $\mathbf{X} = \mathbf{L}+\mathbf{S}$ in a similar manner. However, the difference is that $\mathbf{S}$ in (19) contains only sparse signals, whereas $\mathbf{S}$ in RPCA is a noise matrix.

As noted above, both the randomized power iteration algorithm [31] and GoDec [16] provide a means of overcoming the singular values with slow decay operating on $\tilde{\boldsymbol{X}} = (\mathbf{X}\mathbf{X}^T)^{\gamma}\mathbf{X}$ rather than the original input matrix $\mathbf{X}$. In a randomized power iteration algorithm, it uses $\tilde{\mathbf{X}}$ to perform random sampling and then to obtain the sampling matrix. On the other hand, GoDec uses BRP to solve the following low-rank approximation:

$$\mathbf{L} = \mathbf{Q}_1\left[\mathbf{R}_1\left(\mathbf{A}_2^T\mathbf{Y}_1\right)^{-1}\mathbf{R}_2^T\right]^{1/(2\gamma+1)}\mathbf{Q}_2^T \tag{20}$$

where $\mathbf{Q}_1$, $\mathbf{Q}_2$, $\mathbf{R}_1$, $\mathbf{R}_2$ are QR decomposition of $\mathbf{Y}_1$, $\mathbf{Y}_2$. In addition, the model in (19) can be further solved by minimizing the following decomposition error:

$$\begin{aligned} &\min_{\mathbf{L}_{N\times L}, \mathbf{S}_{N\times L}} ||\mathbf{X}_{N\times L} - \mathbf{L}_{N\times L} - \mathbf{S}_{N\times L}||_F^2 \\ &\text{s.t. } \text{rank}(\mathbf{L}_{N\times L}) \le r \quad \text{and} \quad \text{cardinality}(\mathbf{S}_{N\times L}) \le jN \end{aligned} \tag{21}$$

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where $||\cdot||_F$ is the Frobenius matrix norm. Since (21) does not have an analytical solution, it must rely on a numerical algorithm, which is GoDec to find solutions by the following iterative processes:

$$\begin{aligned}\mathbf{L}_{N\times L}^{(t)} &= \arg\min_{\text{rank}(\mathbf{L}_{N\times L})\le r} ||\mathbf{X}_{N\times L}-\mathbf{L}_{N\times L}-\mathbf{S}_{N\times L}^{(t-1)}||_F^2\\ \mathbf{S}_{N\times L}^{(t)} &= \arg\min_{\text{cardnality}(\mathbf{S}_{N\times L})\le jN} ||\mathbf{X}_{N\times L}-\mathbf{L}_{N\times L}^{(t-1)}-\mathbf{S}_{N\times L}||_F^2.\end{aligned} \tag{22}$$

The details of implementing GoDec are summarized as follows.

Algorithm 1 GoDec

Input: $\mathbf{X}_{N\times L}$ input data matrix
r: the rank of the low rank matrix $\mathbf{L}$
k: sparsity cardinality for sparsity matrix $\mathbf{S}$
ε : pre-determined the error
γ : power scheme
Output:
$\mathbf{L}_{N\times L}$: Low rank matrix
$\mathbf{S}_{N\times L}$: Sparsity matrix

Initialize: $\mathbf{L}_{N\times L}^{(0)}=\mathbf{X}_{N\times L}$, $\mathbf{S}_{N\times L}^{(0)}=\mathrm{O}_{N\times L}$, $t=0$
While $||\mathbf{X}_{N\times L}-\mathbf{L}_{N\times L}-\mathbf{S}_{N\times L}^{(t-1)}||_F^2/\ \|\mathbf{X}_{N\times L}\|_F^2>\varepsilon$ **Do**
$t=t+1$;
$\tilde{\mathbf{L}}=[(\mathbf{X}_{N\times L}-\mathbf{S}_{N\times L}^{t-1})(\mathbf{X}_{N\times L}-\mathbf{S}_{N\times L}^{t-1})^T]^{\gamma}(\mathbf{X}_{N\times L}-\mathbf{S}_{N\times L}^{t-1})$
$\mathbf{Y}_1=\tilde{\mathbf{L}}\mathbf{A}_1$, $\mathbf{A}_2=\mathbf{Y}_1$;
$\mathbf{Y}_2=(\tilde{\mathbf{L}})^T\mathbf{Y}_1=\mathbf{Q}_2\mathbf{R}_2$, $\mathbf{Y}_1=\tilde{\mathbf{L}}\mathbf{Y}_2=\mathbf{Q}_1\mathbf{R}_1$;
$\mathbf{L}_{N\times L}^{(t)}=\mathbf{Q}_1[\mathbf{R}_1(\mathbf{A}_2^T\mathbf{Y}_1)^{-1}\mathbf{R}_2^T]^{1/(2\gamma+1)}\mathbf{Q}_2^T$;

$$\mathbf{S}_{N\times L}^{(t)}=P_{\Omega}(\mathbf{X}_{N\times L}-\mathbf{L}_{N\times L}^{(t)}) \tag{23}$$

where Ω is the nonzero subset of the first k largest entries of $|\mathbf{X}_{N\times L}-\mathbf{L}_{N\times L}^{(t)}|$;
End While

It should be noted that the four parameters, r, k, ε, and γ, in the above algorithm must be determined in advance as initial conditions. Since there are no guidelines provided in [16], these parameters have been selected empirically by experiments. Fortunately, in a recent work [30], Chang *et al.* interpreted and rederived GoDec as an OSP-based GoDec (OSP-GoDec) that only required determining the parameter k. In particular, this OSP-GoDec took advantage of VD in [1], [32], [33], [34], [35], and [36] and the MX-SVD developed in [37] to automatically determine the value of k. This significant benefit allows GoDec to be implemented without determining all four parameters, r, k, ε, and γ empirically. Thus, it makes GoDec more accessible and increases its utilization in many future works to come [14], [15]. The reason that GoDec was used in this is simply because GoDec is widely used in the literature, and the details along with its implementations are available on the website.[1] In this case, those who are interested in BA-TCIMF can easily adapt and implement BA-TCIMF for experiments. Nevertheless, OSP-GoDec can be also used as an alternative to GoDec.

A final remark on the projection P_{Ω} used in (23) is noteworthy. It was derived from video anomaly detection in [16] where SC is specified by a predetermined value k and the space Ω is the set of retaining the k largest nonzero elements in the sparse space $\mathbf{S}$ generated by GoDec. $\mathbf{S}_{N\times L}^{(t)}$ in (23) is simply the space Ω obtained by mapping from $\mathbf{X}_{N\times L}-\mathbf{L}_{N\times L}^{(t-1)}$ onto Ω via the projection P_{Ω}. Various comparative analyses can be found in [14], [15], [17], [30], and [38], specifically, [38]. Those who are interested in scare are encouraged to consult [17], [38] for more details of SC interpretations and derivations In particular, Chang [38] provided how the concept of SC was derived and also offered different versions of SC.

[1]https://sites.google.com/site/godecomposition/home

III. BA-TCIMF

This section presents key ideas that are used to develop BA-TCIMF. Basically, BA-TICMF can be implemented in two successive stages with the global BA in the first stage followed by TBS carried out in the second stage.

A. Global BKG Annihilation

According to the above three approaches, DS, LRaSMD, and CDA described in Section II, three versions of BA-TCIMF, DS-BA-TCIMF, LRaSMD-BA-TCIMF, and CDA-BA-TCIMF, can be derived, each of which is described in detail in as separate subsection.

1) First Stage for BKG Annihilation:

(i) DS-BA-TCIMF spheres data by removing 2OS. The resulting data space is denoted by $\hat{\mathbf{X}}=\mathbf{X}_{\text{DS}}$.
(ii) LRaSMD-BA-TCIMF uses the GoDec-produced low-rank space $\mathbf{L}$ as $\mathbf{U}$ and then annihilates BKG via $P_{\mathbf{L}}^{\perp}$ in (2). The resultant space is denoted by $\mathbf{X}_{\mathbf{L}}^{\perp}$.
(iii) CDA-BA-TCIMF implements the CDA model to produce PC space, **PC,** and applies $P_{\mathbf{PC}}^{\perp}$ in (4) to annihilate BKG. The resultant space is denoted by $\mathbf{X}_{\mathbf{PC}}^{\perp}$.

2) Second Stage for TDE and TBS:

(i) DS-BA-TCIMF extracts BKG signatures from the GoDec-produced low-rank space $\hat{\mathbf{L}}$ in $\hat{\mathbf{X}}=\mathbf{X}_{\text{DS}}$, which are used as additional unwanted signatures to increase and enhance TD. In the meantime, DS-BA-TCIMF also calculates and inverts a sample correlation matrix $\mathbf{R}_{\text{BA}}=\hat{\mathbf{R}}$ in $\hat{\mathbf{X}}=\mathbf{X}_{\text{DS}}$ to improve TBS.
(ii) LRaSMD-BA-TCIMF extracts m BKG signatures from $\mathbf{X}_{\mathbf{L}}$, which are further used as additional unwanted signatures to increase and enhance TD. At the same time, LRaSMD-BA-TCIMF also inverts a sample correlation matrix $\mathbf{R}_{\text{BA}}=\mathbf{R}_{\mathbf{L}+\mathbf{S}}$ obtained from the joint $\mathbf{L}+\mathbf{S}$ space to improve TBS.
(iii) CDA-BA-TCIMF extracts m BKG signatures from $\mathbf{X}_{\mathbf{PC}}$, which are used as additional unwanted signatures to increase and enhance TD. Then, it also inverts a sample correlation matrix $\mathbf{R}_{\text{BA}}=\mathbf{R}_{\mathbf{PC}+\mathbf{IC}}$ calculated from the joint $\mathbf{PC}+\mathbf{IC}$ space to improve TBS.

B. Target BKG Annihilation

After the global BKG annihilation, we further look into TBS. More specifically, we want to enhance TD by annihilating additional BKG signatures extracted from the BA data space while inverting the BA correlation matrix, denoted by $\mathbf{R}_{\text{BA}}$ to achieve further BS. In doing so, TCIMF divided targets of interest into desired targets, $\mathbf{d}_1,\mathbf{d}_2,\ldots,\mathbf{d}_p$, and undesired targets, $\mathbf{u}_1,\mathbf{u}_2,\ldots,\mathbf{u}_q$, embedded in BKG. Since TCIMF is

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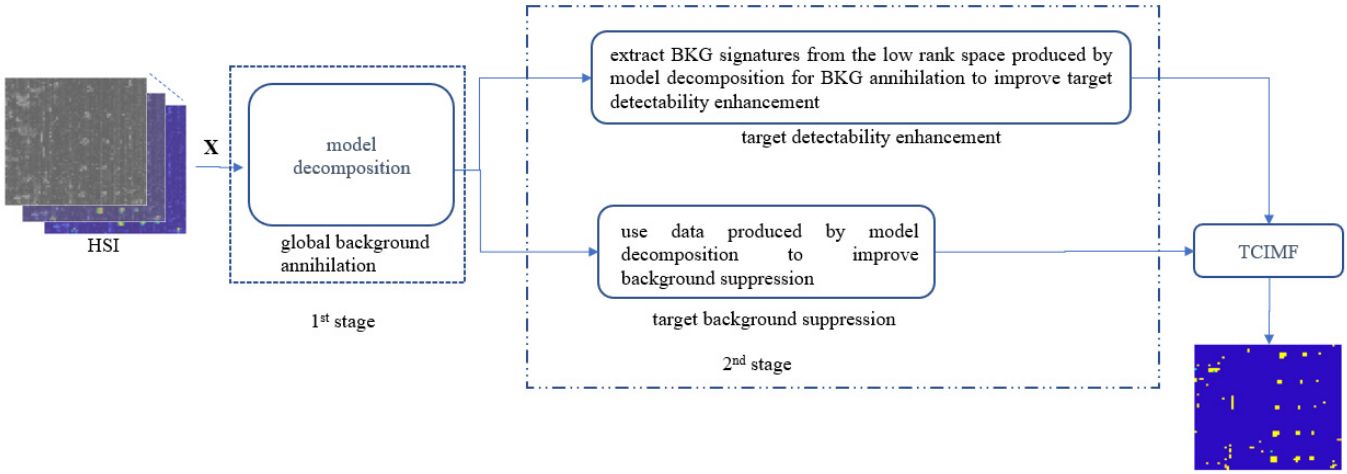


Fig. 1. Two stages implemented in BA-TCIMF.

very sensitive to its use of target knowledge via OSP, the contamination of BKG effect on these targets may result in poor detection performance. Thus, TCIMF can be implemented in two consecutive stages with the first stage to annihilate global BKG prior to acquiring target knowledge and then followed by the second stage to extract $\mathbf{D}$ and $\mathbf{U}$ from this annihilated BKG, denoted by $\mathbf{D}_{\mathrm{BA}}$ and $\mathbf{U}_{\mathrm{BA}}$. Apparently, there are also unknown BKG signatures that cannot be identified by visual inspection or prior knowledge and should be included in $\mathbf{U}_{\mathrm{BA}}$ for further BKG annihilation to enhance TD. In doing so, the singular value decomposition (SVD) can be used for this purpose to extract additional m BKG signatures, $\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_m$ to form a BKG signature matrix $\mathbf{B}_m = [\mathbf{b}_1 \mathbf{b}_2, \ldots, \mathbf{b}_m]$ that can be implemented in conjunction with $\mathbf{D}_{\mathrm{BA}}\mathbf{U}_{\mathrm{BA}}$ to replace $\mathbf{DU}$ in (1) as $\mathbf{D}_{\mathrm{BA}}\mathbf{U}_{\mathrm{BA}}\mathbf{B}_m$. Such signature augmentation is similar to the SDIN model used by SDIA but offers better BKG annihilation than SDIA because its knowledge is obtained from the annihilated BKG not from the original data space. In the meantime, TCIMF also takes advantage of CEM to suppress target BKG via inverting a correlation matrix of annihilated BKG, $\mathbf{R}_{\mathrm{BA}}$, to replace the global $\mathbf{R}$ in (2) to improve target BS. Accordingly, there are two key processes that improve TCIMF: global BKG annihilation in the first stage followed by the second stage that includes additional BKG signatures to augment the BA $\mathbf{D}_{\mathrm{BA}}\mathbf{U}_{\mathrm{BA}}$ for better BKG annihilation, so as to enhance TD, while also improving target BS by using the annihilated correlation matrix $\mathbf{R}_{\mathrm{BA}}$. This leads to a new improved two-stage TCIMF, called BA-TCIMF, as shown in Fig. 1, where the first stage is to annihilate BKG prior to TCIMF followed by the second stage that performs both TDE and TBS.

Depending upon how to perform global BKG annihilation for BA-TCIMF, three versions of BA-TCIMF can be derived as follows.

1) DS-BA-TCIMF: The first version of BA-TCIMF is to use DS to annihilate the 2OS-characterized global BKG prior to TCIMF. It is then followed by applying GoDec to the sphered data $\hat{\mathbf{X}}$ to yield the sphered LRaSMD representation given by

$$\hat{\mathbf{X}} = \hat{\mathbf{L}} + \hat{\mathbf{S}} + \hat{\mathbf{N}}. \tag{24}$$

Using $\hat{\mathbf{L}}$ in (18), MX-SVD is used to find m BKG signatures $\{\hat{\mathbf{b}}_1, \hat{\mathbf{b}}_2, \ldots, \hat{\mathbf{b}}_m\}$ to form a BKG signature matrix $\hat{\mathbf{B}}_{\hat{\mathbf{L}}} = [\hat{\mathbf{b}}_1 \hat{\mathbf{b}}_2, \ldots, \hat{\mathbf{b}}_m]$, i.e., $\hat{\mathbf{B}}_{\hat{\mathbf{L}}} = \mathbf{B}_m$ in Fig. 2(a). By coupling with $\hat{\mathbf{D}}\hat{\mathbf{U}} = [\hat{\mathbf{d}}_1 \hat{\mathbf{d}}_2, \ldots, \hat{\mathbf{d}}_p, \hat{\mathbf{u}}_1 \hat{\mathbf{u}}_2, \ldots, \hat{\mathbf{u}}_q]$ obtained from the $\mathbf{DU}$ in the sphered data $\hat{\mathbf{X}} = \mathbf{X}_{\mathrm{DS}}$, we can augment $\mathbf{DU}$ in (1) to a new target matrix $\hat{\mathbf{D}}\hat{\mathbf{U}}\hat{\mathbf{B}} = [\hat{\mathbf{d}}_1 \hat{\mathbf{d}}_2, \ldots, \hat{\mathbf{d}}_p, \hat{\mathbf{u}}_1 \hat{\mathbf{u}}_2, \ldots, \hat{\mathbf{u}}_q, \hat{\mathbf{b}}_1 \hat{\mathbf{b}}_2, \ldots, \hat{\mathbf{b}}_m]$ to yield a new constraint matrix

$$\left(\hat{\mathbf{D}}\hat{\mathbf{U}}\hat{\mathbf{B}}\right)^T \hat{\mathbf{w}} = \begin{bmatrix} \mathbf{1}_{p\times 1} \\ \mathbf{0}_{q\times 1} \\ \mathbf{0}_{m\times 1} \end{bmatrix} = \left(\underbrace{1, \ldots, 1}_{p}, \underbrace{0, \ldots, 0}_{q}, \underbrace{0, \ldots, 0}_{m}\right)^T \tag{25}$$

for further BKG annihilation to enhance TD. In the meantime, $\mathbf{R}$ in (2) is replaced by $\hat{\mathbf{R}} = (1/N)\sum_{i=1}^{N} \hat{\mathbf{r}}_i \hat{\mathbf{r}}_i^T$, which is the correlation matrix formed by data samples in $\hat{\mathbf{X}}$ to further improve target BS in $\hat{\mathbf{X}}$. By virtue of $\hat{\mathbf{D}}\hat{\mathbf{U}}\hat{\mathbf{B}}$ in (25) and $\hat{\mathbf{R}}$, the constrained optimization problem (2) can be extended to a sphered target-augmented constrained optimization problem given by

$$\min_{\hat{\mathbf{w}}}\left\{\hat{\mathbf{w}}^T \hat{\mathbf{R}} \hat{\mathbf{w}}\right\} \quad \text{s.t.} \quad \left(\hat{\mathbf{D}}\hat{\mathbf{U}}\hat{\mathbf{B}}\right)^T \hat{\mathbf{w}} = \begin{bmatrix} \mathbf{1}_{p\times 1} \\ \mathbf{0}_{q\times 1} \\ \mathbf{0}_{m\times 1} \end{bmatrix}. \tag{26}$$

The optimal solution to (26) is the weight vector $(\hat{\mathbf{w}})^*$ given by

$$(\hat{\mathbf{w}})^* = \hat{\mathbf{R}}^{-1}\hat{\mathbf{D}}\hat{\mathbf{U}}\hat{\mathbf{B}}\left(\left(\hat{\mathbf{D}}\hat{\mathbf{U}}\hat{\mathbf{B}}\right)^T \hat{\mathbf{R}}^{-1}\hat{\mathbf{D}}\hat{\mathbf{U}}\hat{\mathbf{B}}\right)^{-1}\begin{bmatrix} \mathbf{1}_{p\times 1} \\ \mathbf{0}_{q\times 1} \\ \mathbf{0}_{m\times 1} \end{bmatrix} \tag{27}$$

which yields the following DS-BA-TCIMF:

$$\delta^{\text{DS-BA-TCIMF}}(\mathbf{r}) = \left((\hat{\mathbf{w}})^*\right)^T \hat{\mathbf{r}}. \tag{28}$$

An algorithm to implement DS-BA-TCIMF depicted in Fig. 2(a) is summarized in the following.

It should be noted that the OSP-target detection developed in [14] turns out to be a special case of DS-BA-TCIMF.

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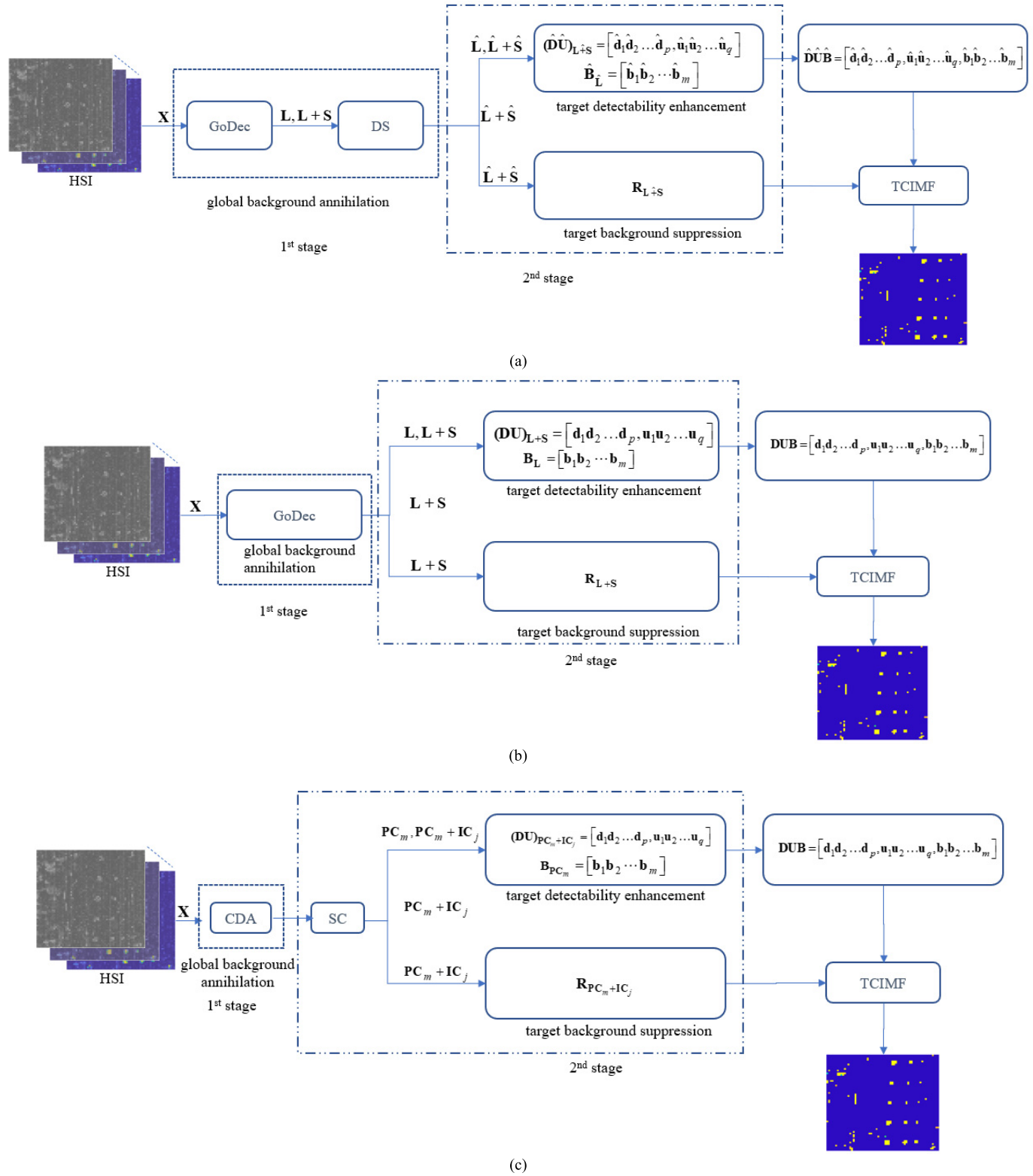


Fig. 2. Diagrams of three versions of BA-TCIMF. (a) Diagram of DS-BA-TCIMF implemented in two stages. (b) Diagram of LRaSMD-BA-TCIMF implemented in two stages. (c) Diagram of CDA-BA-TCIMF implemented in two stages.

2) LRaSMD-BA-TCIMF: The LRaSMD model developed in [16] also offers a feasible solution to resolving TBS arising in TCIMF. It decomposes a data space into the low-rank space **L**, sparse space **S**, and noise space **N** as (14) where **L** and **S** can be considered as the BKG and target spaces, respectively. Using the OSP specified by (2) can remove BKG, whereas finding an appropriate value of k via SC introduced in GoDec [16] can enhance TD. By including the LSaSMD model in TCIMF, we can derive BA-TCIMF as follows.

Recently, LRaSMD was developed to represent a data space $\mathbf{X}$ as a three-component subspace decomposition, $\mathbf{X} = \mathbf{L} + \mathbf{S} + \mathbf{N}$ with the low-rank space $\mathbf{L}$ and sparse space $\mathbf{S}$

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Algorithm 2 DS-BA-TCIMF

1. Initial condition: given value of m.
2. 1st stage process:
   Sphere $\mathbf{X}$ to obtained sphered data $\hat{\mathbf{X}} = \mathbf{X}_{\mathrm{DS}}$.
3. 2nd stage process:
   a. Implement GoDec on $\hat{\mathbf{X}}$ to produce the
   $$\hat{\mathbf{X}} = \hat{\mathbf{L}} + \hat{\mathbf{S}} + \hat{\mathbf{N}} \tag{29}$$
   b. Apply singular value decomposition (SVD) to $\hat{\mathbf{L}}$ and find m BKG signatures $\{\hat{\mathbf{b}}_1, \hat{\mathbf{b}}_2, \ldots, \hat{\mathbf{b}}_m\}$ from $\hat{\mathbf{L}}$ to form $\hat{\mathbf{B}} = [\hat{\mathbf{b}}_1 \hat{\mathbf{b}}_2 \cdots \hat{\mathbf{b}}_m]$
   c. Form $\hat{\mathbf{D}}\hat{\mathbf{U}}\hat{\mathbf{B}} = [\hat{\mathbf{d}}_1 \hat{\mathbf{d}}_2 \ldots \hat{\mathbf{d}}_p \hat{\mathbf{u}}_1 \hat{\mathbf{u}}_2 \ldots \hat{\mathbf{u}}_q \hat{\mathbf{b}}_1 \hat{\mathbf{b}}_2 \ldots \hat{\mathbf{b}}_m]$.
   d. Replace $\mathbf{R}$ in (15) with $\mathbf{R}_{\mathrm{BA}} = \hat{\mathbf{R}}$ to perform TCIMF.

corresponding to the BKG space and target space respectively, and the noise space is specified by $\mathbf{N}$. By taking advantage of this LRaSMD model, we can make use of OSP to annihilate the BKG prior to TCIMF via $P_{\mathbf{L}}^{\perp}\mathbf{X}$ in (15).

By means of the low-rank space $\mathbf{L}$ generated by the LRaSMD model, we can find additional unwanted BKG signals from $\mathbf{L}$ for further improvement on BS. Using the OSP-GoDec approach proposed in [30], we can find a set of m BKG signals $\{\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_m\}$ to form a BKG matrix $\mathbf{B} = [\mathbf{b}_1 \mathbf{b}_2, \ldots, \mathbf{b}_m]$ so the undesired signature matrix $\mathbf{U}$ in (2) can be further expanded to denote $\mathbf{DUB} = [\mathbf{d}_1 \mathbf{d}_2, \ldots, \mathbf{d}_p, \mathbf{u}_1 \mathbf{u}_2, \ldots, \mathbf{u}_q, \mathbf{b}_1 \mathbf{b}_2, \ldots, \mathbf{b}_m]$. As a result, (2) can be extended and expanded to

$$(\mathbf{DUB})_{\mathbf{L}}^{\perp} = P_{\mathbf{L}}^{\perp}(\mathbf{DUB}). \tag{30}$$

Now, if we let

$$\mathbf{c} = \begin{bmatrix} \mathbf{1}_{p\times 1} \\ \mathbf{0}_{q\times 1} \\ \mathbf{0}_{m\times 1} \end{bmatrix}$$

constrain these three target matrices by

$$\left((\mathbf{DUB})_{\mathbf{L}}^{\perp}\right)^T \mathbf{w} = \begin{bmatrix} \mathbf{1}_{p\times 1} \\ \mathbf{0}_{q\times 1} \\ \mathbf{0}_{m\times 1} \end{bmatrix} = \left(\underbrace{1, \ldots, 1}_{p}, \underbrace{0, \ldots, 0}_{q}, \underbrace{0, \ldots, 0}_{m} \right)^T \tag{31}$$

where $\mathbf{DUB} = [\mathbf{d}_1 \mathbf{d}_2, \ldots, \mathbf{d}_p \mathbf{u}_1 \mathbf{u}_2, \ldots, \mathbf{u}_q \mathbf{b}_1 \mathbf{b}_2, \ldots, \mathbf{b}_m]$, $\mathbf{1}_{p\times 1}$ is a $p \times 1$ vector with ones in all components, $\mathbf{0}_{q\times 1}$ is a $q \times 1$ vector with zero in its components, and $\mathbf{0}_{m\times 1}$ is a $m \times 1$ vector with zero in its components, where 1 represents the desired targets and 0 represents the undesired targets. Using (31) as a constraint vector, (2) can be custom-designed as the following constrained optimization problem:

$$\min_{\mathbf{w}} \{\mathbf{w}^T \mathbf{R}_{\mathbf{L}+\mathbf{S}} \mathbf{w}\} \quad \text{s.t.} \quad \left((\mathbf{DUB})_{\mathbf{L}}^{\perp}\right)^T \mathbf{w} = \begin{bmatrix} \mathbf{1}_{p\times 1} \\ \mathbf{0}_{q\times 1} \\ \mathbf{0}_{m\times 1} \end{bmatrix} \tag{32}$$

where the correlation matrix $\mathbf{R}$ in (2) is replaced with $\mathbf{R}_{\mathbf{L}+\mathbf{S}}$ formed by data samples in the $\mathbf{L} + \mathbf{S}$ space. The optimal solution to (20) is the weight vector $\left(\mathbf{w}_{\mathbf{L}}^{\perp}\right)^*$ given by

$$\left(\mathbf{w}_{\mathbf{L}}^{\perp}\right)^* = \mathbf{R}_{\mathbf{L}+\mathbf{S}}^{-1}(\mathbf{DUB})_{\mathbf{L}}^{\perp} \left(\left((\mathbf{DUB})_{\mathbf{L}}^{\perp}\right)^T \mathbf{R}_{\mathbf{L}+\mathbf{S}}^{-1}(\mathbf{DUB})_{\mathbf{L}}^{\perp} \right)^{-1} \begin{bmatrix} \mathbf{1}_{p\times 1} \\ \mathbf{0}_{q\times 1} \\ \mathbf{0}_{m\times 1} \end{bmatrix}. \tag{33}$$

The FIR filter specified by $\left(\mathbf{w}_{\mathbf{L}}^{\perp}\right)^*$ in (33) is called the BA-TCIMF given by

$$\delta^{\text{LRaSMD-BA-TCIMF}}(\mathbf{r}) = \left(\mathbf{w}_{\mathbf{L}}^{\perp}\right)^* P_{\mathbf{L}}^{\perp}\mathbf{r}. \tag{34}$$

An algorithm to implement LRaSMD-BA-TCIMF described in Fig. 2(b) can be summarized in the following.

Algorithm 3 LRaSMD-BA-TCIMF

1. Initial condition: given value of m.
2. 1st stage process:
   a. Implement DoDec to produce
   $$\mathbf{X} = \mathbf{L} + \mathbf{S} + \mathbf{N} \tag{35}$$
   b. Apply $P_{\mathbf{L}}^{\perp}$ in (2) to remove $\mathbf{L}$ for BKG annihilation.
3. 2nd stage process:
   a. Apply SVD to $\mathbf{L}$ to find m BKG signatures $\{\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_m\}$ from $\mathbf{L}$ to form $\mathbf{B}_{\mathbf{L}} = [\mathbf{b}_1 \mathbf{b}_2 \cdots \mathbf{b}_m]$.
   b. Form $(\mathbf{DU})_{\mathbf{L}+\mathbf{S}}\mathbf{B}_{\mathbf{L}}$ and implement (20) with $\mathbf{DUB}$ replaced by $(\mathbf{DU})_{\mathbf{L}+\mathbf{S}}\mathbf{B}_{\mathbf{L}}$.
   c. Replace $\mathbf{R}$ in (15) with $\mathbf{R}_{\mathrm{BA}} = \mathbf{R}_{\mathbf{L}+\mathbf{S}}$ which is the correlation matrix formed by data samples in the joint $\mathbf{L} + \mathbf{S}$ space respectively to perform TCIMF.

3) CDA-BA-TCIMF: As an alternative to LRaSMD-BA-TCIMF, a CDA-based BA-TCIMF can be also derived in a similar manner. It replaces the LRaSMD model in (14) with the CDA model in (16) proposed by Chen *et al.* [17]

$$\mathbf{X} = \mathbf{PC}_m + \mathbf{IC}_j + \mathbf{N} \tag{36}$$

with $\mathbf{PC}_m$ and $\mathbf{IC}_j$ corresponding to $\mathbf{L}$ and $\mathbf{S}$, respectively. As a result of (36), a CDA-BA-TCIMF can be readily derived with its algorithm described in the following.

Algorithm 4 CDA-BA-TCIMF

1. Initial condition: given value of m.
2. 1st stage:
   a. Implement CDA on $\mathbf{X}$ to produce (30).
   b. Apply $P_{\mathbf{PC}_m}^{\perp}$ in (4) to remove $\mathbf{PC}_m$ space for BKG annihilation.
3. 2nd stage:
   a. Implement SC to retain sparse signal sources. The resulting space is referred to as CDASC space.
   b. Find the first m eigenvectors from the CDASC space as BKG signatures $\{\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_m\}$ from $\mathbf{PC}_m$ to form $\mathbf{B}_{\mathbf{PC}_m} = [\mathbf{b}_1 \mathbf{b}_2 \cdots \mathbf{b}_m]$.
   c. Form $\mathbf{D}_{\mathrm{c}}\mathbf{U}_{\mathrm{c}}\mathbf{B}_{\mathbf{PC}_m}$ and implement (20) with $\mathbf{DUB}$ replaced by $(\mathbf{D}_{\mathrm{c}}\mathbf{U}_{\mathrm{c}})_{\mathbf{PC}_m}\mathbf{B}_{\mathbf{PC}_m}$
   d. Replace $\mathbf{R}$ in (15) with $\mathbf{R}_{\mathrm{BA}} = \mathbf{R}_{\mathbf{PC}_m+\mathbf{IC}_j}$ which is the correlation matrix formed by data sample the joint $\mathbf{PC}_m + \mathbf{IC}_j$ respectively to perform TCIMF.

Fig. 2(c) depicts the diagram of CDA-BA-TCIMF where BKG is annihilated by CDA.

IV. Image Data and Detection Measures

A. HYDICE Data Scene

Fig. 3 shows that the image scene to be studied was captured in August 1995 by the aerial hyperspectral digital imagery collection experiment (HYDICE) sensor from a flight altitude of 10 000 ft. This scene has been investigated in depth by many reports, such as [1], [2], [3], and [4]. It has a size of $\mathbf{IC}_j$ pixel vectors, where low signal/high noise bands: bands

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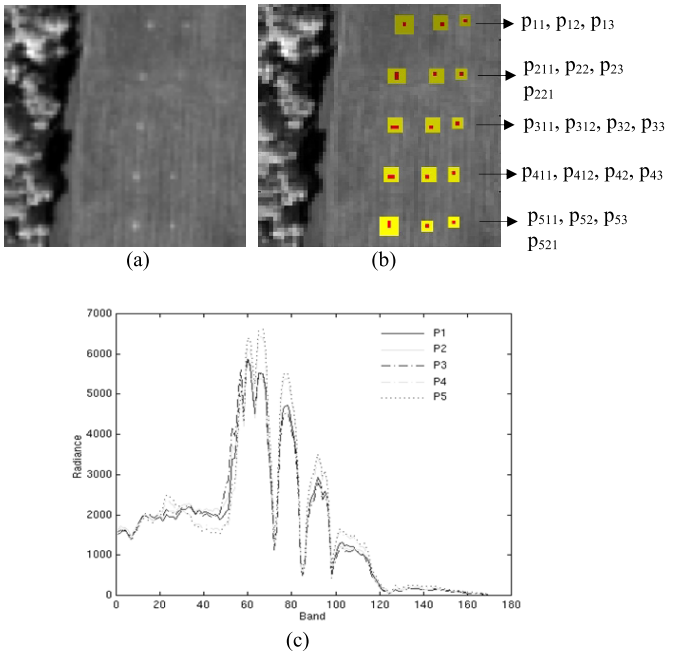


Fig. 3. (a) HYDICE panel scene that contains 15 panels. (b) Ground-truth map of spatial locations of 19 R panel pixels. (c) Spectra of $\mathbf{p}_1$, $\mathbf{p}_2$, $\mathbf{p}_3$, $\mathbf{p}_4$, and $\mathbf{p}_5$.

TABLE I

PARAMETERS OF p AND j ESTIMATED BY NWHFC USING MX-SVD FOR HYDICE AND SAN DIEGO AIRPORT DATASETS AND PARAMETER SETTINGS FOR KTCIMF

estimation of p and j by NWHFC using MX-SVD			
Data	VD	p	j by MX-SVD
HYDCE 15-panel scene	NWHFC $P_F \leq 10^{-3}$	13	6
San Diego airport scene	HFC, $P_F \leq 10^{-3}$	10	2
parameter setting for KTCIMF			
Data	kernel	standard deviation	
HYDCE 15-panel scene	Gaussian	4.92×10^{-11}	
San Diego airport scene	Gaussian	5.88×10^{-12}	

1–3 and bands 202–210 and water vapor absorption bands: bands 101–112 and bands 137–153 were removed to yield 169 spectral bands. The spatial resolution is 1.56 m, and the spectral resolution is 10 nm. The scene is consisting of 15 square panels with three different sizes, 3 m × 3 m, 2 m × 2 m, and 1 m × 1 m, respectively, as shown in Fig. 3(a). Since the ground sampling distance is approximately 1.56 m, the panels in the first column except the first row have two-panel pixels that are highlighted by red: p_{211} and p_{221} in row 2, p_{311} and p_{312} in row 3, p_{411} and p_{412} in row 4, and p_{511} and p_{521} in row 5, as shown in Fig. 3(b), in the ground truth. All the remaining 11 panels in Fig. 3(b) only contain one single panel pixel for each panel also highlighted by red: p_{11}, p_{12}, and p_{13} in row 1, p_{22} and p_{23} in row 2, p_{32} and p_{33} in row 3, p_{42} and p_{43} in row 4, and p_{52} and p_{53} in row 5. Adding them up, the scene has a total of 19-panel pixels. Their precise spatial locations are shown in Fig. 3(b) with the pixels in yellow (Y pixels) indicating panel pixels mixed with the BKG. The five-panel spectral signatures are plotted in Fig. 3(c), $\mathbf{p}_1$, $\mathbf{p}_2$, $\mathbf{p}_3$, $\mathbf{p}_4$, and $\mathbf{p}_5$, which were obtained from Fig. 3(b), where $\mathbf{p}_i$ is the ith panel signature and was calculated by averaging the red panel pixels in row i.

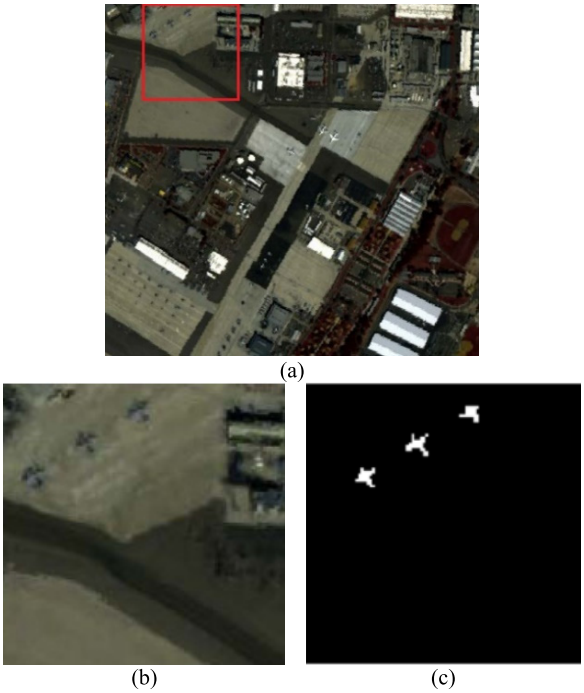


Fig. 4. AVIRIS San Diego airport scene. (a) Pseudocolor image of the whole scene. (b) Pseudocolor image of the selected area. (c) Ground-truth map.

This particular scene was used for subpixel target detection of panel pixels in the third column and mixed target detection of panel pixels highlighted by yellow in the first and second columns. Detailed descriptions of this data scene and discussions on experiments can be found in [1, Sects. 9.3 and 9.4], [2], and [3].

One major key issue in implementing GoDec is the determinization of the values of r and k, which are the order of low-rank subspace $\mathbf{L}$ and the sparsity of a sparse subspace $\mathbf{S}$, respectively. The OSP-GoDec proposed in [1], [19], [20], [21], [22], and [23] used VD and MX-SVD that were derived from the maximum orthogonal complement algorithm (MOCA) developed in [23] to resolve this issue. Instead of directly estimating r and k, it first used VD to estimate a parameter p, which is assumed to be the number of spectral signatures. The value of p can be used to determine the sum of the order of $\mathbf{L}$, denoted by m and the rank of $\mathbf{S}$, denoted by j as $p = m + j$. As a result, r and k used in GoDec can be obtained by letting $r = m = p - j$ and $k = j \times N$. In the following experiments, the same idea of using VD/MX-SVD was also used to estimate m as r and $k = j \times N$ for GoDec.

B. AVIRIS San Diego Airport Scene

Fig. 4(a) shows a second data scene to be studied is an AVIRIS airport scene from San Diego, CA, USA. The size of the airport is 400 × 400 pixels, and its spatial resolution is 3.5 m with 224 spectral bands, which is the third dimensionality ranging from 370 to 2510 nm. Some bands corresponding to the water absorption regions, the low signal-to-noise ratio (SNR), and collapse bands (1–6, 33–35, 97, 107–113, 153–166, and 221–224) have been removed and, finally, results in 189 bands. A red square was marked at the upper left corner of the scene that was chosen as the test image with its

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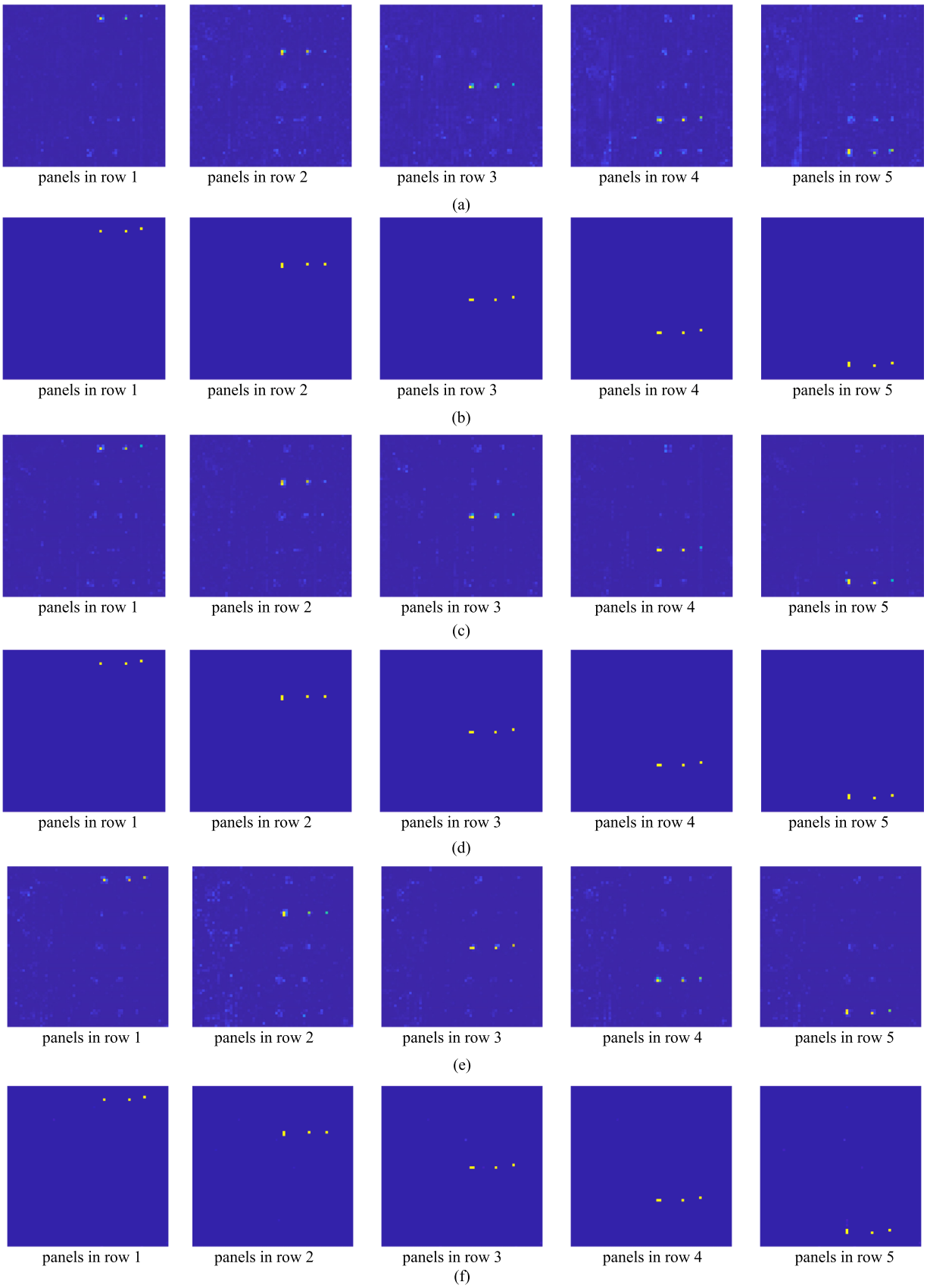


Fig. 5. Target abundance detection maps of various target detectors for HYDICE data scene. (a) Detection maps of DS-BA-TCIMF. (b) Detection maps of DS-BA-KTCIMF. (c) Detection maps of LRaSMD-BA-TCIMF. (d) Detection maps of LRaSMD-BA-KTCIMF. (e) Detection maps of CDASC-BA-TCIMF. (f) Detection maps of CDASC-BA-KTCIMF. (g) Detection maps of OSP. (h) Detection maps of CEM. (i) Detection maps of TCIMF. (j) Detection maps of KTCIMF. (k) Detection maps of SDIA. (l) Detection maps of OSPTD. (m) Detection maps of HCEM. (n) Detection maps of ECEM.

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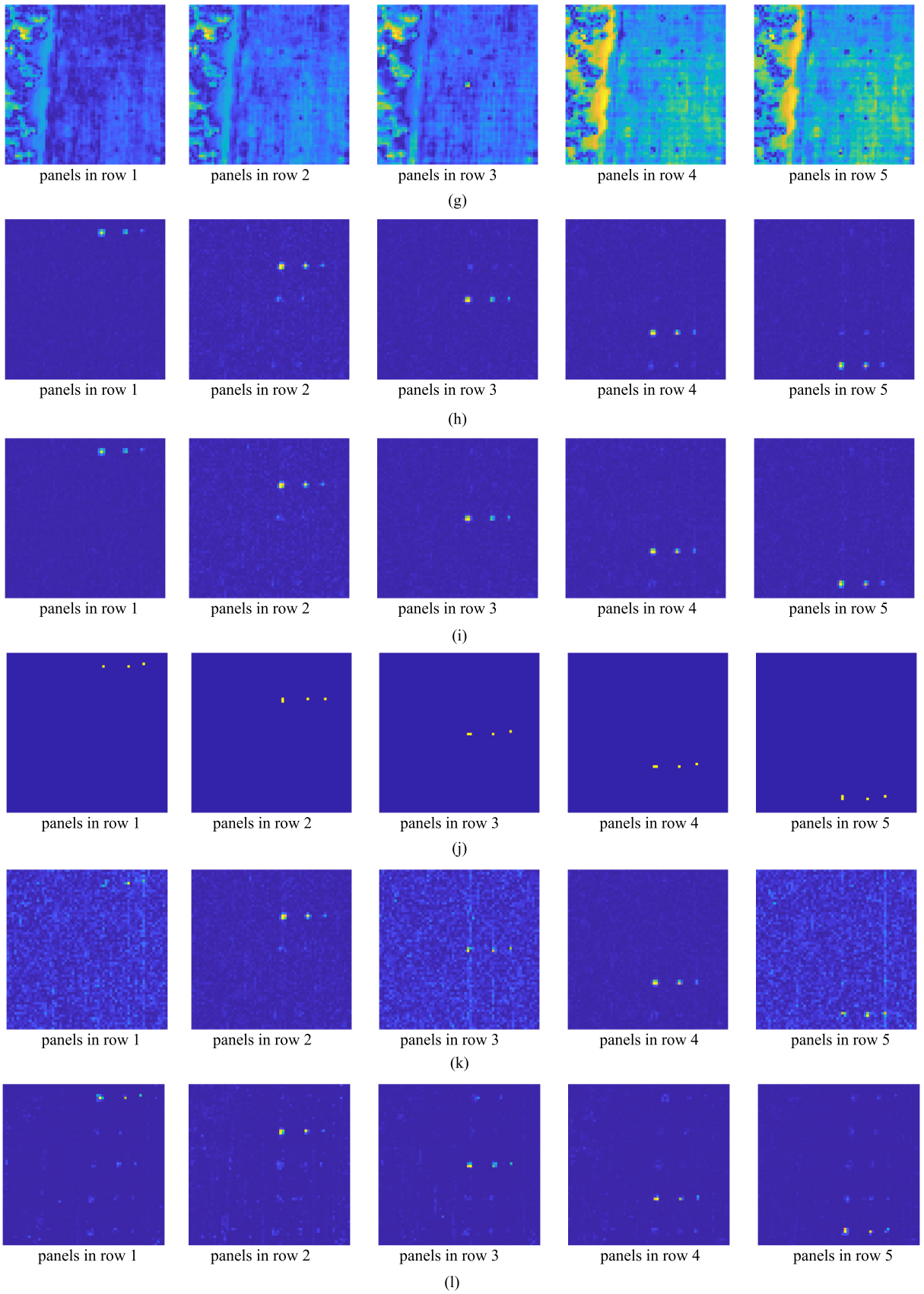


Fig. 5. *(Continued.)* Target abundance detection maps of various target detectors for HYDICE data scene. (a) Detection maps of DS-BA-TCIMF. (b) Detection maps of DS-BA-KTCIMF. (c) Detection maps of LRaSMD-BA-TCIMF. (d) Detection maps of LRaSMD-BA-KTCIMF. (e) Detection maps of CDASC-BA-TCIMF. (f) Detection maps of CDASC-BA-KTCIMF. (g) Detection maps of OSP. (h) Detection maps of CEM. (i) Detection maps of TCIMF. (j) Detection maps of KTCIMF. (k) Detection maps of SDIA. (l) Detection maps of OSPTD. (m) Detection maps of HCEM. (n) Detection maps of ECEM.

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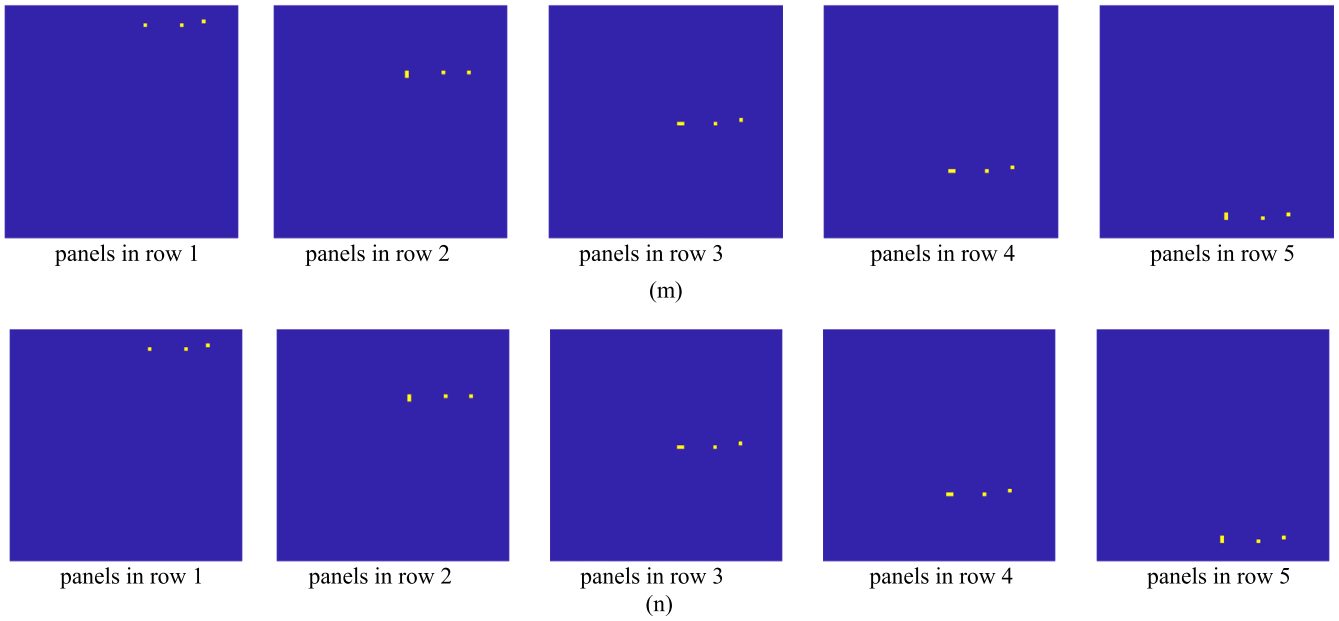


Fig. 5. *(Continued.)* Target abundance detection maps of various target detectors for HYDICE data scene. (a) Detection maps of DS-BA-TCIMF. (b) Detection maps of DS-BA-KTCIMF. (c) Detection maps of LRaSMD-BA-TCIMF. (d) Detection maps of LRaSMD-BA-KTCIMF. (e) Detection maps of CDASC-BA-TCIMF. (f) Detection maps of CDASC-BA-KTCIMF. (g) Detection maps of OSP. (h) Detection maps of CEM. (i) Detection maps of TCIMF. (j) Detection maps of KTCIMF. (k) Detection maps of SDIA. (l) Detection maps of OSPTD. (m) Detection maps of HCEM. (n) Detection maps of ECEM.

ground-truth image shown in Fig. 4(b) and (c), respectively. The restricted area has a size of 100 × 100 pixels. The BKG of the scene is mainly made up of a roof, shadow, and grass. The targets that we are interested in are three airplanes with the same materials in the scene, which comprised 85 pixels and account for 0.33% of the test image, as shown in Fig. 4(b). The location of these planes can be obtained from the ground truth in Fig. 4(c) and then be used as the desired signature $\mathbf{d}$.

To estimate VD, HFC and its noise-whitened HFC (NWHFC) [1], [19], [38] were used to estimate p. This was followed by MX-SVD to estimate j. Table I tabulates the value of p and j estimated by $P_F \leq 10^{-3}$ and MX-SVD so that $m = p - j$ and $k = j \times N$ and parameter setting for KTCIMF. In the following experiments, γ was set to 2 [16], and $\varepsilon = 10^{-3}$. It should be noted that the parameter q used in GoDec to address the singular value decay problem is now replaced by γ because q is used by TCIMF in (25) as the number of undesired target signatures.

It should be noted that, in the following experiments, the four parameters, r, k, ε, and γ, used in the original GoDec were implemented by letting r to $m = p - j$ and γ to 2. The experiments for different settings of j and p have been studied in great detail in [14], [15], [17], and [30]. The values used in this article were selected from the best cases reported in these references.

In addition, the computer environment used for experiments is documented in Table II.

TABLE II
COMPUTER ENVIRONMENT

computer environment parameter	configuration
OS Name:	Microsoft Windows 10 Home
OS Version:	10.0.19043 N/A Build 19043
OS Manufacturer:	Microsoft Corporation
System Type:	x64-based PC
Processor(s):	Intel(R) Core(TM) i7-8750H CPU @ 2.20GHz 2.21 GHz
Total Physical Memory:	32,624 MB
Graphics Card:	GeForce GTX 1060 Graphics Cards

C. Target Detection Measures

1) Targets' Detection Measure: In [26], it is noted that $\text{AUC}_{(D,F)}$ cannot fully evaluate a detector, and it can be only used to evaluate the effectiveness of a detector. By taking the factor $\text{AUC}_{(D,\tau)}$ into $\text{AUC}_{(D,F)}$ account, then a new TD measure that derives from both the 2-D ROC curve of (P_D, P_F) and the 2-D ROC curve of (P_D, τ), denoted by AUC_{TD} as follows, is established:

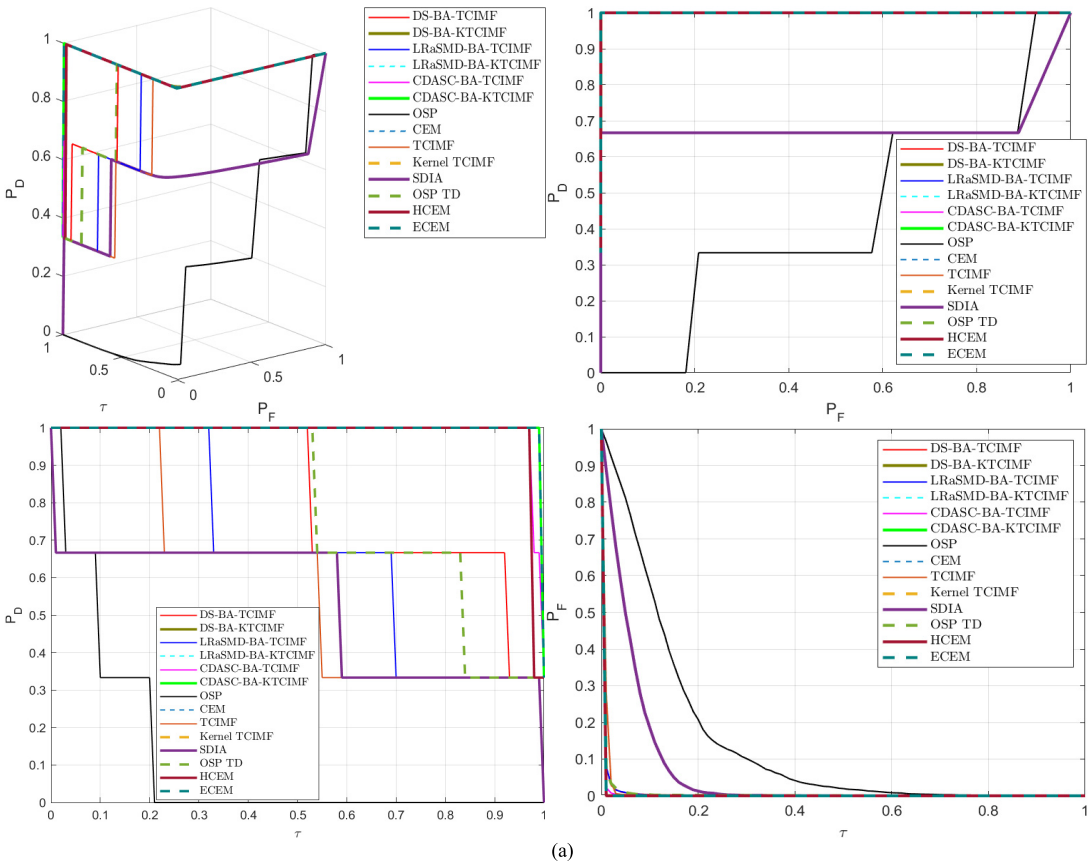
$$0 \leq \text{AUC}_{\text{TD}} = \text{AUC}_{(D,F)} + \text{AUC}_{(D,\tau)}) \leq 2. \tag{37}$$

2) Joint Target Detection BKG Suppression Measure: In the Neyman–Pearson detection theory, it maximizes P_D by fixing P_F. Therefore, P_D and P_F are two critical factors to evaluate the performance of a detector. Just using P_D to solely evaluate the detection performance is not enough and reasonable since BKG has a tremendous impact on target detection and worth paying extra consideration in detecting targets where the targets and BKG are mixed and be embedded at the subpixel level. Thus, to measure the performance of BS in hyperspectral target detection, P_F is introduced. However, current measurements used P_D and P_F separately, to measure a detector more comprehensively, using a joint TDBS measurement are more suitable than using P_D and P_F solely and separately. Taking P_F into different considerations, two detection measures related to P_F can be derived as follows.

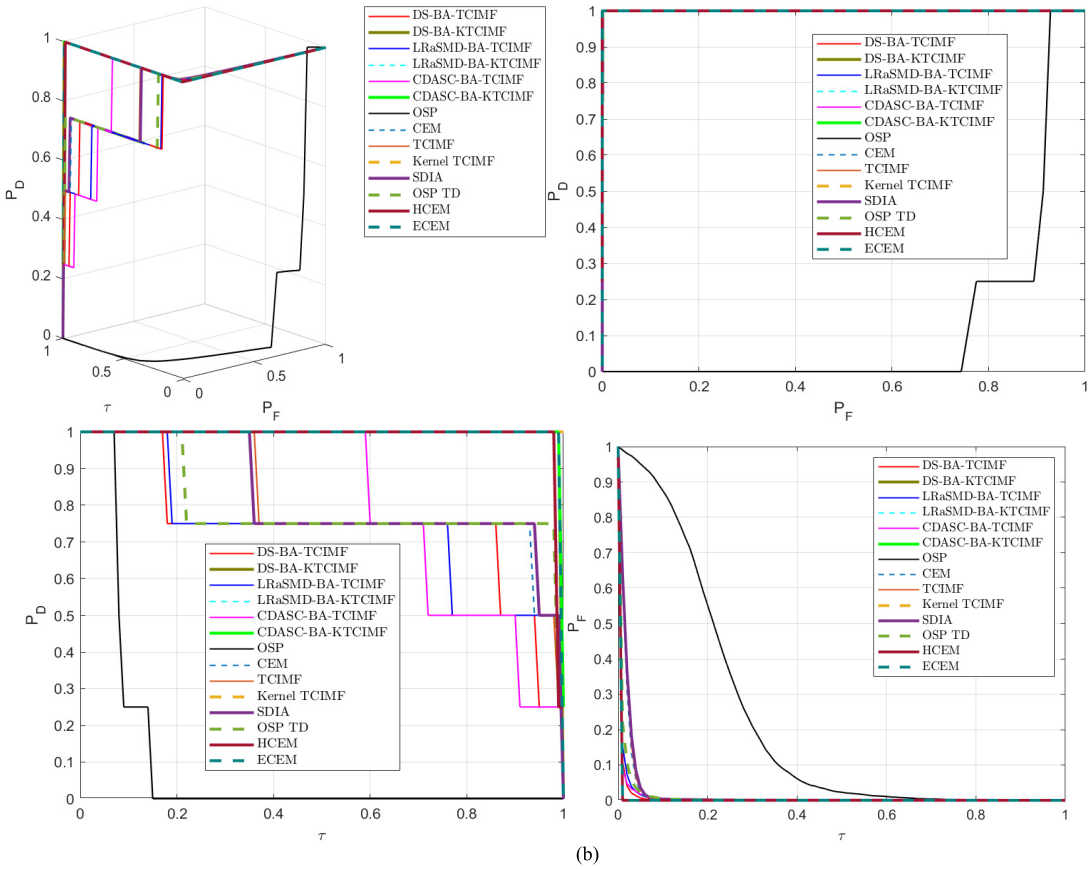
One of them is called the BS measure defined by

$$-1 \leq \text{AUC}_{\text{BS}} = \text{AUC}_{(D,F)} - \text{AUC}_{(F,\tau)}) \leq 1. \tag{38}$$

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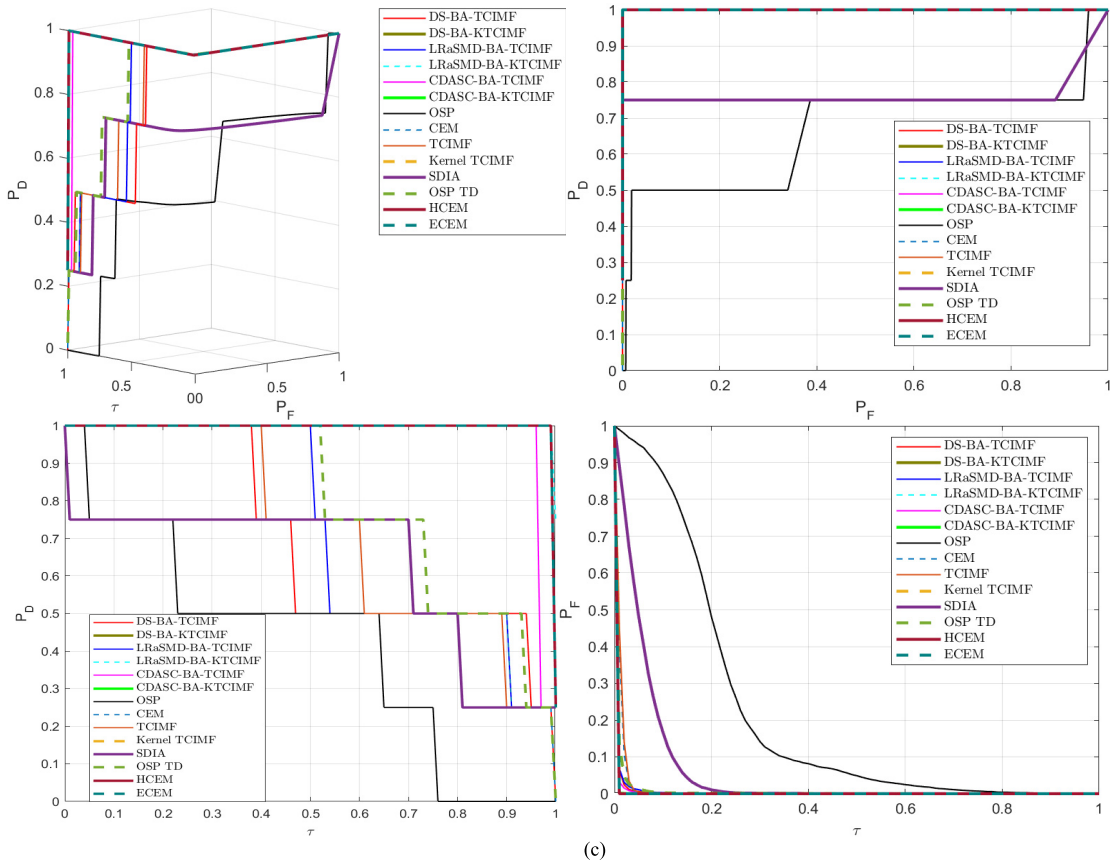
(a)



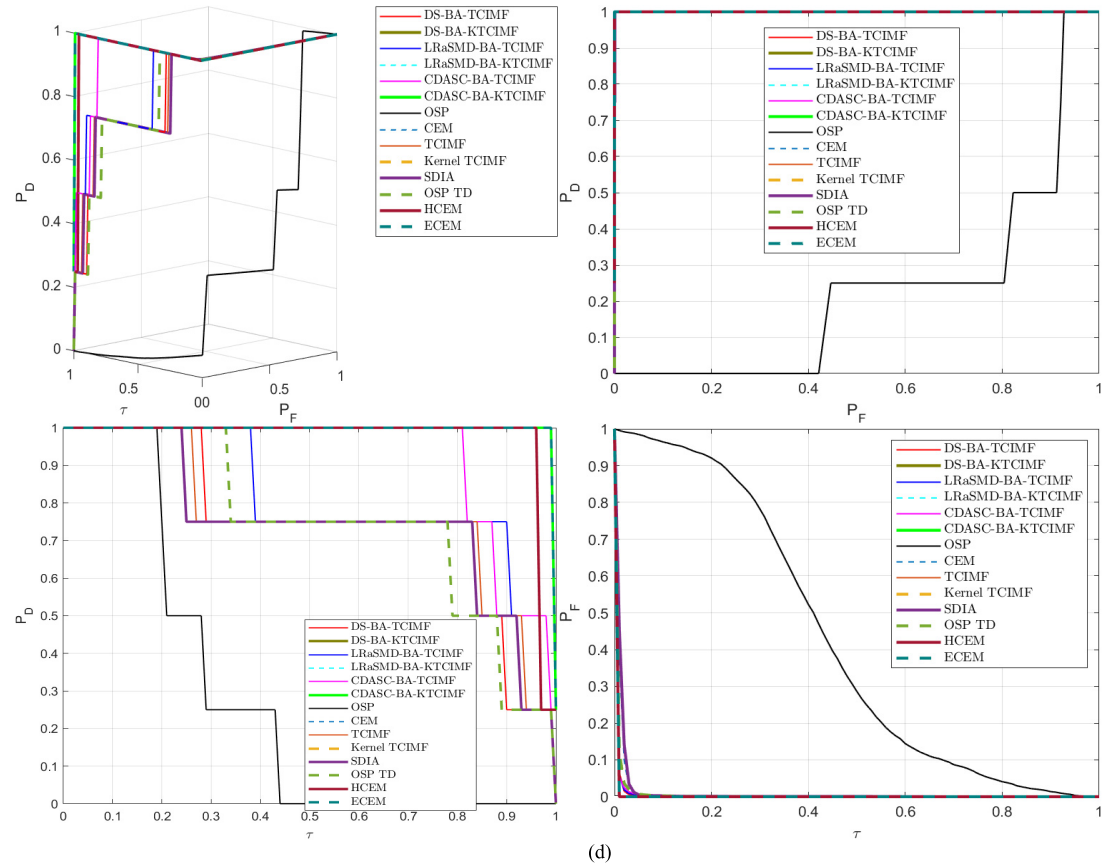
(b)

Fig. 6. ROC curves generated from the results in Fig. 5 for the HYDICE data scene. (a) 3-D ROC curves and 2-D ROC curves for P_1 in the HYDICE data scene. (b) 3-D ROC curves and 2-D ROC curves for P_2 in the HYDICE data scene. (c) 3-D ROC curves and 2-D ROC curves for P_3 in the HYDICE data scene. (d) 3-D ROC curves and 2-D ROC curves for P_4 in the HYDICE data scene. (e) 3-D ROC curves and 2-D ROC curves for P_5 in the HYDICE data scene.

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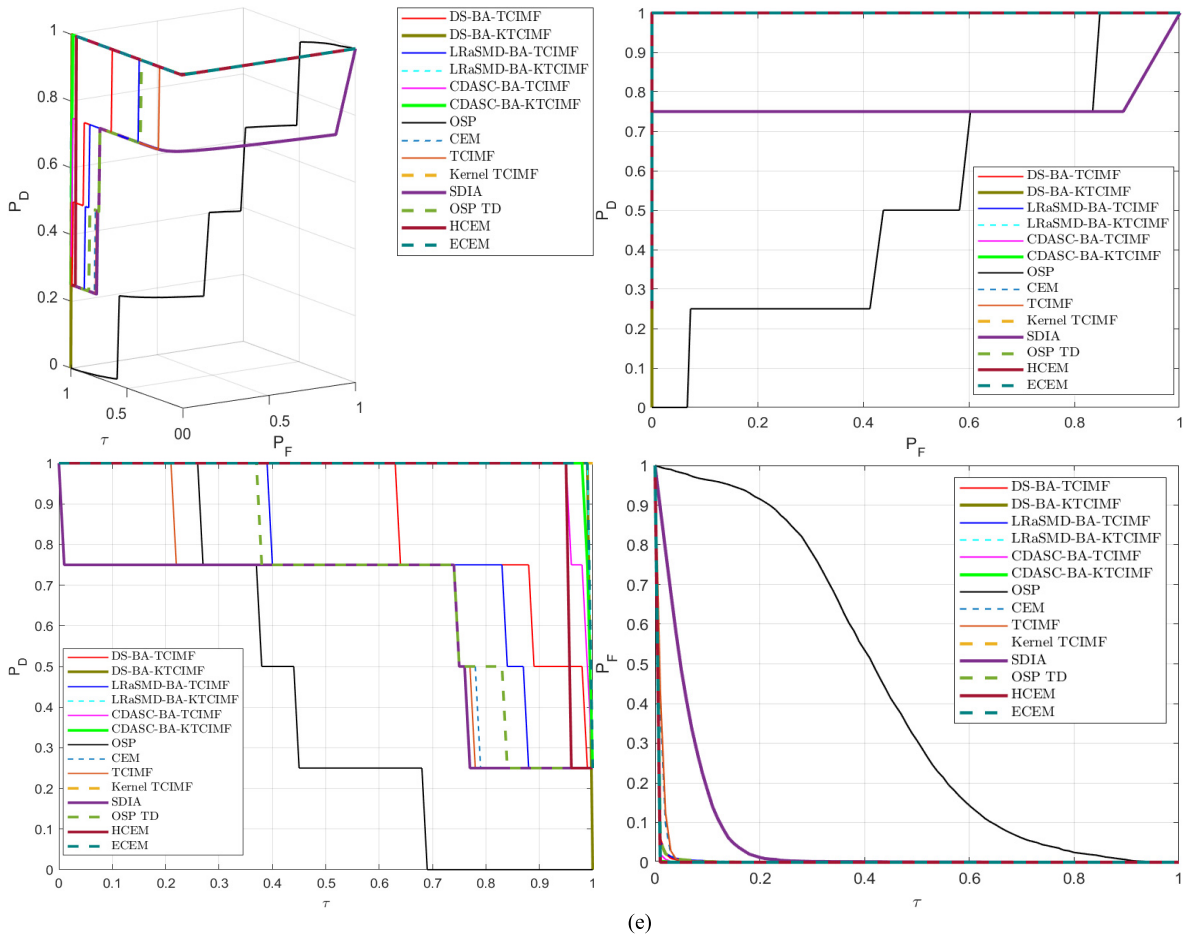
(c)



(d)

Fig. 6. *(Continued.)* ROC curves generated from the results in Fig. 5 for the HYDICE data scene. (a) 3-D ROC curves and 2-D ROC curves for P_1 in the HYDICE data scene. (b) 3-D ROC curves and 2-D ROC curves for P_2 in the HYDICE data scene. (c) 3-D ROC curves and 2-D ROC curves for P_3 in the HYDICE data scene. (d) 3-D ROC curves and 2-D ROC curves for P_4 in the HYDICE data scene. (e) 3-D ROC curves and 2-D ROC curves for P_5 in the HYDICE data scene.

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(e)

Fig. 6. *(Continued.)* ROC curves generated from the results in Fig. 5 for the HYDICE data scene. (a) 3-D ROC curves and 2-D ROC curves for P_1 in the HYDICE data scene. (b) 3-D ROC curves and 2-D ROC curves for P_2 in the HYDICE data scene. (c) 3-D ROC curves and 2-D ROC curves for P_3 in the HYDICE data scene. (d) 3-D ROC curves and 2-D ROC curves for P_4 in the HYDICE data scene. (e) 3-D ROC curves and 2-D ROC curves for P_5 in the HYDICE data scene.

It measures the effectiveness of detectors by taking P_F into consideration carefully.

The other is TDBS defined by

$$-1 \leq \text{AUC}_{\text{TDBS}} = \text{AUC}_{(D,\tau)} - \text{AUC}_{(F,\tau)}) \leq 1. \tag{39}$$

It measures the target detection ability by taking P_F into consideration carefully.

3) Signal-to-Noise Probability Ratio: The idea of SNR, which is extensively used in communication, can also be utilized to create a detection measure termed SNPR, similar to AUC_{TDBS} (SNPR), denoted by AUC_{SNPR}, as

$$0 \leq \text{AUC}_{\text{SNPR}} = \frac{\text{AUC}_{(D,\tau)}}{\text{AUC}_{(F,\tau)}} \tag{40}$$

where $\text{AUC}_{(D,\tau)}$ can be viewed as signals and $\text{AUC}_{(F,\tau)}$ can be viewed as noise.

There is a significant difference between SNR and SNPR. SNR is widely used in communications and signal processing, which calculates the ratio of signal energy to noise energy. SNPR calculates the ratio of $\text{AUC}_{(D,\tau)}$ to $\text{AUC}_{(F,\tau)}$, both of which are probabilities where $\text{AUC}_{(D,\tau)}$ and $\text{AUC}_{(F,\tau)}$ correspond to target detection probability solely determined by the threshold τ and false alarm probability also solely determined by the threshold τ, respectively.

4) Overall Detection Measure: As you may be noticed that TD in (37), BS in (38), TDBS in (39), and SNPR in (40) all only considered the two 2-D ROC curves, this section derives a new detection measure by taking into account AUC values of its corresponding three 2-D ROC curves, 2-D ROC curves of (P_D, P_F), 2-D ROC curves of (P_D, τ), and 2-D ROC curves of (P_F, τ), named overall detection probability (ODP), denoted by AUC_{ODP}, as

$$-1 \leq \text{AUC}_{\text{ODP}} = \text{AUC}_{(D,F)} + \text{AUC}_{(D,\tau)} - \text{AUC}_{(F,\tau)} \leq 2. \tag{41}$$

This detection measure tends to measure how effectiveness and detection performance of a detector by considering the impact of BS. The reason is that higher values of $\text{AUC}_{(D,\text{F})}$ and $\text{AUC}_{(D,\tau)}$ mean higher detection performance. Conversely, a lower value of $\text{AUC}_{(F,\tau)}$ indicates a better BS and, thus, a better detection performance. As a result, (41) combines the AUC values obtained by the three 2-D ROC curves to provide a single quantitative measurement of overall detection performance.

Because $\text{AUC}_{(F,\tau)}$ indicates false alarm probability, $1 - \text{AUC}_{(F,\tau)}$ is actually no-signal predictive probability in [19] or no-signal detectability. As a result, we may develop a new detection measure by replacing $\text{AUC}_{(F,\tau)}$ with

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TABLE III
AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES FOR FIVE ROWS DETECTION RESULTS

d	detector	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{SNPR}	AUC_{ODP}	Time(s)
p_1	DS-BA-TCIMF	**1.0000**	0.6900	0.0085	1.6900	0.9915	0.6815	80.8639	1.6815	3.3254
	DS-BA-KTCIMF	**1.0000**	**0.9967**	**0.0050**	**1.9967**	**0.9950**	**0.9917**	**199.3333**	**1.9917**	283.5625
	LRaSMD-BA-TCIMF	**1.0000**	0.6267	**0.0068**	1.6267	0.9932	0.6199	92.4472	1.6199	2.9029
	LRaSMD-BA-KTCIMF	**1.0000**	**0.9967**	**0.0050**	**1.9967**	**0.9950**	**0.9917**	**199.3333**	**1.9917**	125.1055
	CDASC-BA-TCIMF	**1.0000**	**0.9900**	**0.0053**	**1.9900**	**0.9947**	**0.9847**	185.1528	**1.9847**	0.8918
	CDASC-BA-KTCIMF	**1.0000**	**0.9967**	**0.0050**	1.9967	**0.9950**	**0.9917**	**198.8475**	**1.9917**	105.7600
	OSP [5]	0.4333	0.1083	0.1430	0.5416	0.2903	-0.0347	0.7574	0.3986	1.3042
	CEM [7-8]	**0.9993**	0.5883	0.0090	1.5876	0.9903	0.5793	65.3030	1.5786	1.1518
	TCIMF [10]	**0.9993**	0.5883	0.0090	1.5876	0.9903	0.5793	65.4628	1.5786	**0.7553**
	KTCIMF [18]	**1.0000**	**1.0000**	**0.0050**	**2.0000**	**0.9950**	**0.9950**	**200.0000**	**1.9950**	45.7554
	SDIA [12]	**0.6852**	0.5283	0.0603	1.2135	0.6248	0.4680	8.7576	1.1532	**0.7694**
	OSPTD [14]	**1.0000**	0.6633	**0.0068**	1.6633	**0.9932**	0.6566	98.2458	1.6566	3.1729
	HCEM [21]	**1.0000**	0.9833	**0.0050**	1.9833	**0.9950**	0.9783	196.6667	1.9783	**0.7807**
	ECEM [22]	**1.0000**	**0.9967**	**0.0050**	**1.9967**	**0.9950**	**0.9917**	**199.3333**	**1.9917**	15.6134
p_2	DS-BA-TCIMF	**1.0000**	0.8588	0.0108	1.8588	0.9892	0.8479	79.3946	1.8479	3.1457
	DS-BA-KTCIMF	**1.0000**	**0.9963**	**0.0050**	1.9963	**0.9950**	**0.9913**	**199.2500**	**1.9913**	282.2028
	LRaSMD-BA-TCIMF	**1.0000**	0.7438	0.0075	1.7438	**0.9925**	0.7362	98.9410	1.7362	2.7293
	LRaSMD-BA-KTCIMF	**1.0000**	**0.9963**	**0.0050**	1.9963	**0.9950**	**0.9913**	**199.2500**	**1.9913**	127.1160
	CDASC-BA-TCIMF	**1.0000**	**0.9738**	0.0057	**1.9738**	**0.9943**	0.9681	172.1203	1.9681	0.7122
	CDASC-BA-KTCIMF	**1.0000**	**0.9963**	**0.0050**	1.9963	**0.9950**	**0.9912**	**197.9920**	**1.9912**	125.3686
	OSP [5]	0.6643	0.4175	0.2162	1.0818	0.4480	0.2013	1.9307	0.8655	1.4318
	CEM [7-8]	**0.9997**	0.7275	0.0112	1.7272	0.9885	0.7163	64.8427	1.7160	1.2721
	TCIMF [10]	**0.9997**	0.7263	0.0120	1.7259	0.9877	0.7142	60.3537	1.7139	**0.5656**
	KTCIMF [18]	**1.0000**	**1.0000**	**0.0050**	**2.0000**	**0.9950**	**0.9950**	**200.0000**	**1.9950**	44.5655
	SDIA [12]	**0.7636**	0.6288	0.0582	1.3923	0.7053	0.5705	10.7980	1.3341	**0.5977**
	OSPTD [14]	**1.0000**	0.7550	**0.0072**	1.7550	0.9928	0.7478	104.9766	1.7478	3.0110
	HCEM [21]	**1.0000**	**0.9963**	**0.0050**	**1.9963**	**0.9950**	**0.9913**	**199.2500**	**1.9913**	**0.5979**
	ECEM [22]	**1.0000**	**0.9963**	**0.0050**	**1.9963**	**0.9950**	**0.9913**	**199.2500**	**1.9913**	15.6970
p_3	DS-BA-TCIMF	**1.0000**	0.8588	0.0108	1.8588	0.9892	0.8479	79.3946	1.8479	3.1443
	DS-BA-KTCIMF	**1.0000**	**0.9963**	**0.0050**	**1.9963**	**0.9950**	0.9913	**199.2500**	**1.9913**	283.1608
	LRaSMD-BA-TCIMF	**1.0000**	0.7438	0.0075	1.7438	**0.9925**	0.7362	98.9410	1.7362	2.7206
	LRaSMD-BA-KTCIMF	**1.0000**	**0.9963**	**0.0050**	**1.9963**	**0.9950**	**0.9913**	**199.2500**	**1.9913**	126.7084
	CDASC-BA-TCIMF	**1.0000**	**0.9738**	**0.0057**	**1.9738**	**0.9943**	0.9681	172.1203	1.9681	0.7051
	CDASC-BA-KTCIMF	**1.0000**	**0.9963**	**0.0050**	**1.9963**	**0.9950**	**0.9912**	**197.9920**	**1.9912**	115.3024
	OSP [5]	0.6643	0.4175	0.2162	1.0818	0.4480	0.2013	1.9307	0.8655	1.5684
	CEM [7-8]	**0.9997**	0.7275	0.0112	1.7272	0.9885	0.7163	64.8427	1.7160	1.3834
	TCIMF [10]	**0.9997**	0.7263	0.0120	1.7259	0.9877	0.7142	60.3537	1.7139	**0.5664**
	KTCIMF [18]	**1.0000**	**1.0000**	**0.0050**	**2.0000**	**0.9950**	**0.9950**	**200.0000**	**1.9950**	47.9064
	SDIA [12]	**0.7636**	0.6288	0.0582	1.3923	0.7053	0.5705	10.7980	1.3341	0.5823
	OSPTD [14]	**1.0000**	0.7550	**0.0072**	1.7550	0.9928	0.7478	104.9766	1.7478	3.0006
	HCEM [21]	**1.0000**	**0.9963**	**0.0050**	**1.9963**	**0.9950**	**0.9913**	**199.2500**	**1.9913**	**0.5819**
	ECEM [22]	**1.0000**	**0.9963**	**0.0050**	**1.9963**	**0.9950**	**0.9913**	**199.2500**	**1.9913**	15.6800
p_4	DS-BA-TCIMF	**1.0000**	0.8063	0.0091	1.8063	0.9909	0.7971	88.3077	1.7971	3.1312
	DS-BA-KTCIMF	**1.0000**	**0.9950**	**0.0050**	**1.9950**	**0.9950**	**0.9900**	**199.0000**	**1.9900**	283.4540
	LRaSMD-BA-TCIMF	**1.0000**	0.7863	**0.0065**	1.7863	**0.9935**	0.7797	120.2293	1.7797	2.7018
	LRaSMD-BA-KTCIMF	**1.0000**	**0.9963**	**0.0050**	**1.9963**	**0.9950**	**0.9913**	**199.2500**	**1.9913**	127.6413
	CDASC-BA-TCIMF	**1.0000**	0.9188	**0.0062**	1.9188	**0.9938**	0.9126	148.5391	1.9126	0.6925
	CDASC-BA-KTCIMF	**1.0000**	**0.9963**	**0.0050**	**1.9963**	**0.9950**	**0.9912**	**199.1527**	**1.9912**	108.1894
	OSP [5]	0.2280	0.2800	0.4259	0.5080	-0.1978	-0.1459	0.6575	0.0822	1.6673
	CEM [7-8]	**0.9998**	0.7613	0.0109	1.7611	0.9889	0.7504	69.9222	1.7502	1.4795
	TCIMF [10]	**0.9998**	0.7600	0.0117	1.7598	0.9881	0.7483	64.7226	1.7481	**0.5583**
	KTCIMF [18]	**1.0000**	**0.9963**	**0.0050**	**1.9963**	**0.9950**	**0.9913**	**199.2500**	**1.9913**	46.1595
	SDIA [12]	**0.9998**	0.7500	0.0121	1.7498	0.9877	0.7379	61.9000	1.7377	**0.5713**
	OSPTD [14]	**0.9999**	0.7263	0.0066	1.7262	0.9933	0.7196	109.5398	1.7196	2.9864

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TABLE III
(Continued.) AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES FOR FIVE ROWS DETECTION RESULTS

	HCEM [21]	**1.0000**	**0.9738**	**0.0050**	**1.9738**	**0.9950**	0.9688	194.7500	1.9688	**0.5698**
	ECEM [22]	**1.0000**	**0.9963**	**0.0050**	**1.9963**	**0.9950**	**0.9913**	**199.2500**	**1.9913**	15.6463
$\mathbf{p}_5$	DS-BA-TCIMF	**1.0000**	0.8800	0.0084	1.8800	0.9916	0.8716	104.9231	1.8716	3.1891
	DS-BA-KTCIMF	**1.0000**	**0.9963**	**0.0050**	1.9963	**0.9950**	**0.9913**	**199.2500**	**1.9913**	285.1055
	LRaSMD-BA-TCIMF	**1.0000**	0.7588	0.0064	1.7588	0.9936	0.7524	119.3697	1.7524	2.7592
	LRaSMD-BA-KTCIMF	**1.0000**	**0.9963**	**0.0050**	1.9963	**0.9950**	**0.9913**	**199.2500**	**1.9913**	127.9219
	CDASC-BA-TCIMF	**1.0000**	0.9838	**0.0053**	1.9838	**0.9947**	0.9784	185.3363	1.9784	0.7475
	CDASC-BA-KTCIMF	**1.0000**	**0.9938**	**0.0050**	**1.9938**	**0.9950**	**0.9887**	**196.9213**	**1.9887**	107.5967
	OSP [5]	0.5176	0.4425	0.4231	0.9601	0.0945	0.0194	1.0458	0.5370	1.8137
	CEM [7-8]	**0.9998**	0.6863	0.0109	1.6860	0.9889	0.6754	63.0333	1.6751	1.6297
	TCIMF [10]	**0.9998**	0.6838	0.0116	1.6835	0.9882	0.6722	59.1023	1.6719	**0.6122**
	KTCIMF [18]	**1.0000**	**1.0000**	**0.0050**	**2.0000**	**0.9950**	**0.9950**	**200.0000**	**1.9950**	45.0478
	SDIA [12]	**0.7634**	0.6288	0.0600	1.3922	0.7035	0.5688	10.4873	1.3322	**0.6242**
	OSPTD [14]	**1.0000**	0.6938	**0.0058**	1.6938	**0.9942**	0.6879	119.4289	1.6879	3.0557
	HCEM [21]	**1.0000**	0.9663	**0.0050**	1.9613	**0.9950**	0.9613	193.2500	1.9613	**0.6214**
	ECEM [22]	**1.0000**	**0.9963**	**0.0050**	**1.9963**	**0.9950**	**0.9913**	**199.2500**	**1.9913**	15.8237

$1 - \mathrm{AUC}_{(F,\tau)}$, denoted by $\mathrm{AUC}_{\mathrm{ODP}}$, as

$$0 \le \mathrm{AUC}_{\mathrm{ODP}} = \mathrm{AUC}_{(D,F)} + \mathrm{AUC}_{(D,\tau)} + (1 - \mathrm{AUC}_{(F,\tau)}) \le 3. \quad (42)$$

This is the same as (41) and can be used to assess a detector's overall performance in terms of effectiveness, TD, and no-signal detectability.

If we believe that signal sources and BKG sources are two distinct groups of sources, used C_{S} to represent the signal class, and used C_{BKG} to indicate the BKG class, along with their class sample ratios determined by $p_{\mathrm{S}} = (n_{\mathrm{S}}/N)$ and $p_{\mathrm{BKG}} = (n_{\mathrm{BKG}}/N)$ respectively, where n_{S}, n_{BKG}, and N represent the number of data samples in signal class, C_{S}, and the BKG class, C_{BKG}, and the total number of data samples. Then, formula (42) can be modified by letting $\mathrm{AUC}_{(D,F)}$ be zero to remove it, and it was reduced to overall accuracy (OA) for a two-class (signal, BKG) classification problem defined by

$$0 \le \mathrm{OA} = p_{\mathrm{S}}\mathrm{AUC}_{(D,\tau)} + p_{\mathrm{BKG}}(1 - \mathrm{AUC}_{(F,\tau)}) \le 1. \quad (43)$$

As a result, formula (43) can be considered a special case of (36).

To facilitate the use of 2-D/3-D ROC analysis, MATLAB-based software codes developed for calculating AUC and plotting ROC are made available at the website of the Remote Sensing Signal and Image Processing Laboratory (RSSIPL), University of Maryland at Baltimore County (UMBC).[2]

V. EXPERIMENTS AND DISCUSSION

This section performs HYDICE and San Diego experiments to evaluate the detection performance of various targeted detectors developed most recently in the literature: TCIMF, CEM, OSPTD implemented by $(\hat{\mathbf{L}}, \hat{\mathbf{L}} + \hat{\mathbf{S}}) - \mathrm{OSP}$ [14], DS-BA-TCIMF, LRaSMD-BA-TCIMF, and hierarchical HCEM [21], and ensemble-based (ECEM) [22]. For the HYDICE experiments, the target $\mathbf{d}$ is specified by one of five-panel signatures in Fig. 3(c), and the remaining four panel signatures will be treated as undesired target signatures, $\mathbf{U}$. For San Diego experiments, since it only has one target specified by the three airplanes in the scene, there is no undesired target signature matrix $\mathbf{U}$. In this case, TCIMF was implemented with $\mathbf{U} = \emptyset$ as an empty set. For OSPTD, DS-BA-TCIMF, and LRaSMD-BA-TCIMFm the sphered low rank and low-rank matrices were used as undesired target signature matrices, respectively, where the low-rank matrix and the sparsity matrix were generated by GoDec described in Section II-C.

A. HYDICE Data Experiments

Fig. 5(a)–(n) shows the detection maps produced by OSP, CEM, SDIA, TCIMF, KTCIMF, OSPTD, DS-BA-TCIMF/KTCIMF, LRaSMD-BA-TCIMF/KTCIMF, HCEM, and ECEM. By visual inspection of Fig. 5, the best results were those produced by BA-KTCIMF where not only the spatial coordinates of 19 R panel pixels were precisely located but also BKG was completed suppressed, as shown in Fig. 5(b), (d), (f), and (j) by BA-KTCIMF, Fig. 5(m) by HCEM, and Fig. 5(n) by ECEM. This was followed by the next best results produced by BA-TICMF without using kernels shown in Fig. 5(a), (c), and (e). Due to the use of the joint matrix of $\mathbf{L} + \mathbf{S}$, the 19 R panel pixels were precisely detected. The five subpixel panels in the third column, which are subpixel, also were identified clearly. What are the most interestingly is the spatial locations of p_{211} and p_{221} that are vertically lined-up in the second row, and the spatial location of p_{311} and p_{312} that are horizontally lined-up in third row, the spatial location of p_{411} and p_{412} that are horizontally lined-up in the fourth row, and the spatial location of p_{511} and p_{521} that are vertically lined-up in the fifth row shown in Fig. 3(b) by ground truth were correctly detected without any blur. The worst result was produced by OSP in Fig. 5(g) followed by the second worst result by SDIA in Fig. 5(k). This mainly due to the fact that panel detection was largely caused by the complicated BKG and the similarity of panels of each row. However, this dilemma was effectively resolved by TCIMF in Fig. 5(i) and further improved by including DS in Fig. 5(a), (c), and (e). In particular, the detection results

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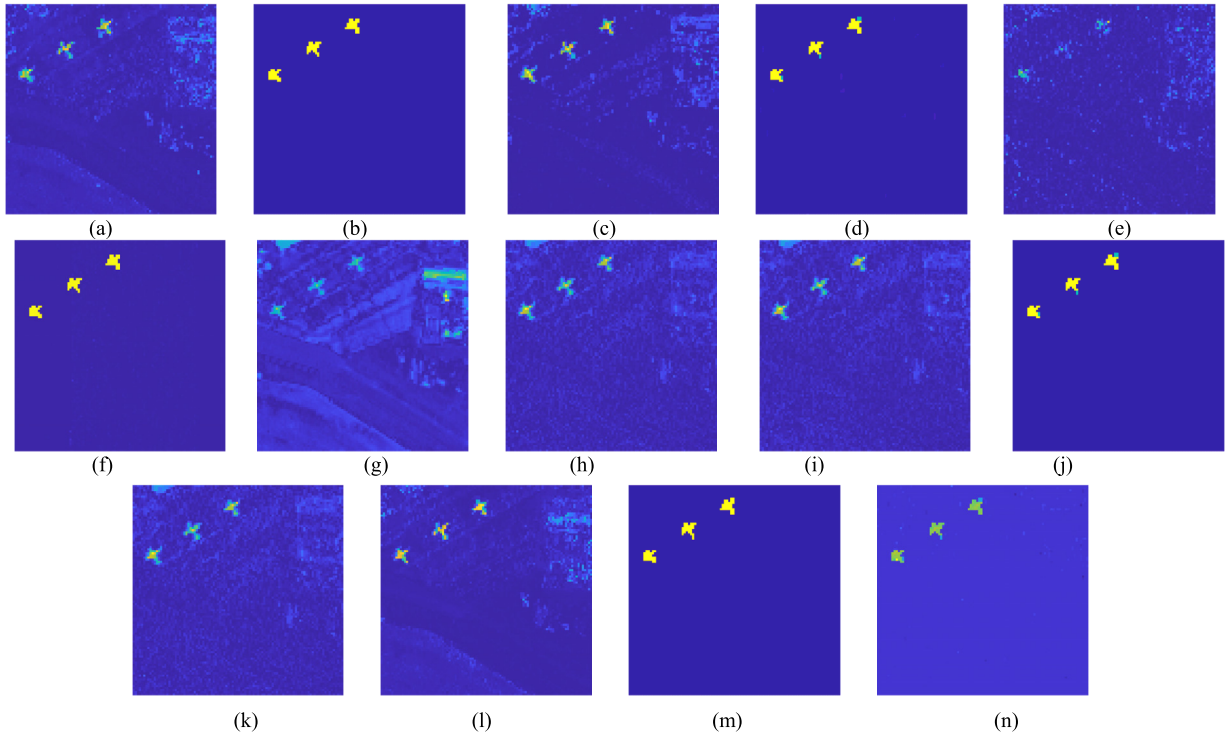


Fig. 7. Target abundance detection maps of various target detectors for the San Diego data scene. (a) DS-BA-TCIMF. (b) DS-BA-KTCIMF. (c) LRaSMD-BA-TCIMF. (d) LRaSMD-BA-KTCIMF. (e) CDASC BA-TCIMF. (f) CDASC-BA-KTCIMF. (g) OSP. (h) CEM. (i) TCIMF. (j) KTCIMF. (k) SDIA. (l) OSP TD. (m) HCEM. (n) ECEM.

by OSP in Fig. 5(g) were amazingly improved by OSPTD in Fig. 5(l) by simply using DS to annihilate BKG prior to panel detection. The results in Fig. 5(a), (c), (e), and (l) indicated that BKG can be largely characterized by first-order statistics and 2OS. Besides, the results in Fig. 5(a), (c), and (e) also showed that BA-TCIMF could be significantly improved by using the low-rank matrix $\mathbf{L}$ as a BKG signature matrix for global BKG annihilation or using $P_{\mathbf{L}}^{\perp}$ to annihilate global BKG. Interestingly, both BA-TCIMF and $(\hat{\mathbf{L}}, \hat{\mathbf{L}} + \hat{\mathbf{S}}) - \text{OSP}$ share the same concept that BKG must be annihilated prior to target detection.

To further conduct quantitative studies and analysis, 3-D ROC curves of the detection results in Fig. 5 along with their corresponding 2-D ROC curves are plotted in Fig. 6 from which the eight detection measures derived in Section IV-C were calculated in Table III for performance evaluation with best in bold, second best in bold red, and third best in bold blue. As also shown in the last column of Table III are the computing times.

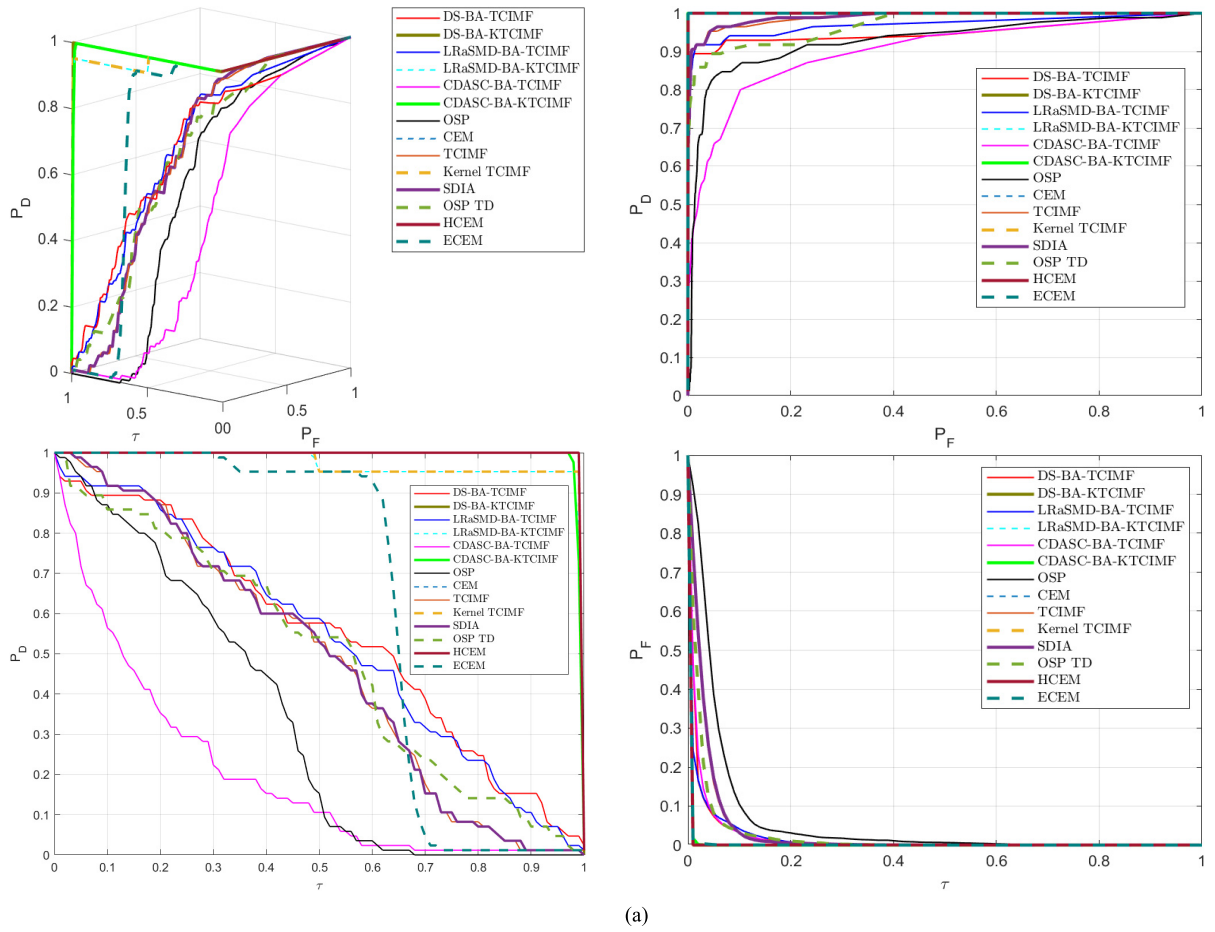
As shown in Table III, all the KTCIMF-based detectors performed extremely well in all detection measures across board and achieved perfect panel detection at the expense of exceedingly high computational complexity. Their corresponding counterpart, BA-TCIMF, produced the second-best results with only inappreciable performance degradation. Thus, if we factor in the issue of computing time, BA-TCIMF is generally preferred to BA-KTCIMF. Also, Table III showed that the best and second best results were very close, but the third best results were little far off from the best results.

It is worth noting from Table III that using the joint matrix of $\mathbf{L} + \mathbf{S}$ and kernels instead of the original data to reduce the influences of noise did improve the panel detection performance dramatically. This can be seen that BA-KTCIMF and BA-TCIMF achieved very high AUC_{SNPR}, specifically BA-KCIMF. In addition, Table III also demonstrated one important fact that relying on $\text{AUC}_{(D,F)}$ is not reliable. For example, according to Table III, both CEM and TCIMF achieved $\text{AUC}_{(D,F)} = 0.9997$ and 0.9998 compared to BA-TCIMF and BA-KTCIMF, which achieved $\text{AUC}_{(D,F)} = 1.000$. The differences between these detectors were within an error range, 2×10^{-4}. With such tiny numerical error tolerance, how can we claim that one is better than another. The 3-D ROC curve-derived detection measures provide such solutions.

B. San Diego Airport Data Experiments

Similar experiments conducted for the HYDICE 15-panel scene were performed for the San Diego airport data scene. Fig. 7 shows the detection maps produced by OSP, CEM, SDIA, TCIMF, KTCIMF, OSPTD, DS-BA-TCIMF/KTCIMF, LRaSMD-BA-TCIMF/KTCIMF, HCEM, and ECEM. In analogy with the HYDICE data, the visual inspection of Fig. 7 showed the best results that were produced by KTCIMF, various versions of BA-KTCIMF, HCEM, and ECEM, all of which had very clean BS. To further conduct quantitative analysis, Fig. 8 plots 3-D ROC curves calculated from the detection maps in Fig. 7 along with their corresponding 2-D ROC curves. Table IV tabulates values of $\text{AUC}_{(D,F)}$, $\text{AUC}_{(D,\tau)}$, $\text{AUC}_{(F,\tau)}$, AUC_{TD}, AUC_{BS}, AUC_{TDBS}, AUC_{SNPR}, and AUC_{ODP} with best in bold, second best in bold red, and third best in bold blue where the best results also confirmed by visual assessment

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(a)

Fig. 8. ROC curves generated from the results in Fig. 7 for the San Diego data scene. 3-D ROC curves and 2-D ROC curves in the San Diego data scene.

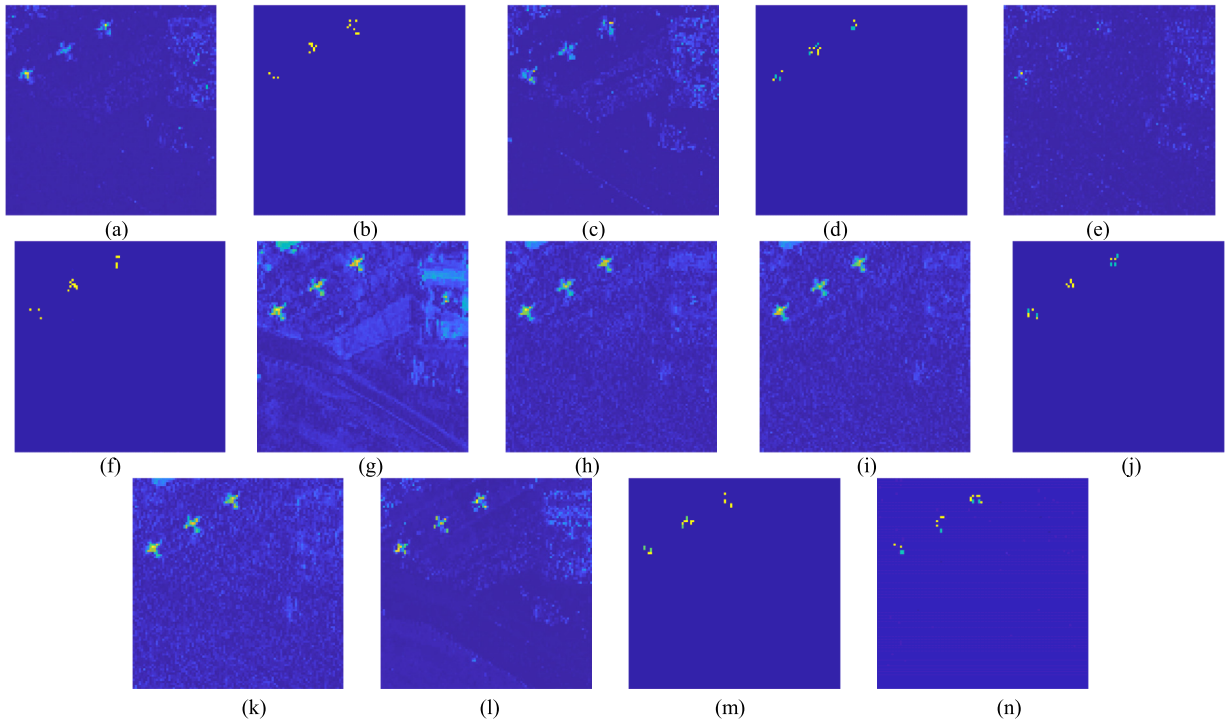


Fig. 9. Detection map of detectors with 20% randomly selected training samples to calculate target signatures. (a) DS-BA-TCIMF. (b) DS-BA-KTCIMF. (c) LRaSMD-BA-TCIMF. (d) LRaSMD-BA-KTCIMF. (e) CDASC-BA-TCIMF. (f) CDASC-BA-KTCIMF. (g) OSP. (h) CEM. (i) TCIMF. (j) KTCIMF. (k) SDIA. (l) OSPTD. (m) HCEM. (n) ECEM.

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TABLE IV
AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES FOR SAN DIEGO DETECTION RESULTS

detector	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{SNPR}	AUC_{ODP}	Time(s)
DS-BA-TCIMF	0.9526	0.5486	0.0183	1.5012	0.9343	0.5303	30.0436	1.4829	9.0906
DS-BA-KTCIMF	**1.0000**	**0.9951**	**0.0050**	**1.9951**	**0.9950**	**0.9901**	**199.0118**	**1.9901**	3298.9927
LRaSMD-BA-TCIMF	0.9637	0.5758	0.0158	1.5394	0.9479	0.5600	36.4111	1.5236	8.4569
LRaSMD-BA-KTCIMF	**1.0000**	**0.9717**	**0.0052**	1.9717	**0.9948**	0.9665	186.0474	1.9665	1420.5184
CDASC-BA-TCIMF	0.9016	0.1949	0.0193	1.0966	0.8824	0.1757	10.1183	1.0773	1.6254
CDASC-BA-KTCIMF	**1.0000**	**0.9921**	**0.0052**	**1.9921**	**0.9948**	**0.9869**	**190.9511**	**1.9869**	1407.2919
OSP [5]	0.9248	0.4036	0.0681	1.3284	0.8567	0.3355	5.9278	1.2603	8.5540
CEM [7-8]	0.9893	0.4819	0.0302	1.4711	0.9590	0.4517	15.9419	1.4409	**1.1488**
TCIMF [10]	0.9877	0.4871	0.0314	1.4748	0.9563	0.4557	15.5122	1.4434	8.0651
KTCIMF [18]	**1.0000**	0.9715	**0.0052**	1.9715	**0.9948**	0.9663	**186.9160**	1.9663	602.8169
SDIA [12]	**0.9903**	0.4826	0.0310	1.4729	0.9593	0.4517	15.5738	1.4419	**1.2361**
OSPTD [14]	0.9687	0.5656	0.0235	1.5342	0.9452	0.5421	24.1180	1.5108	**298.2633**
HCEM [21]	**0.9998**	**0.9951**	**0.0054**	**1.9949**	0.9944	**0.9897**	184.2908	**1.9895**	6.8343
ECEM [22]	**1.0000**	0.6407	**0.0054**	1.6407	**0.9946**	0.6353	118.5297	1.6353	48.4120

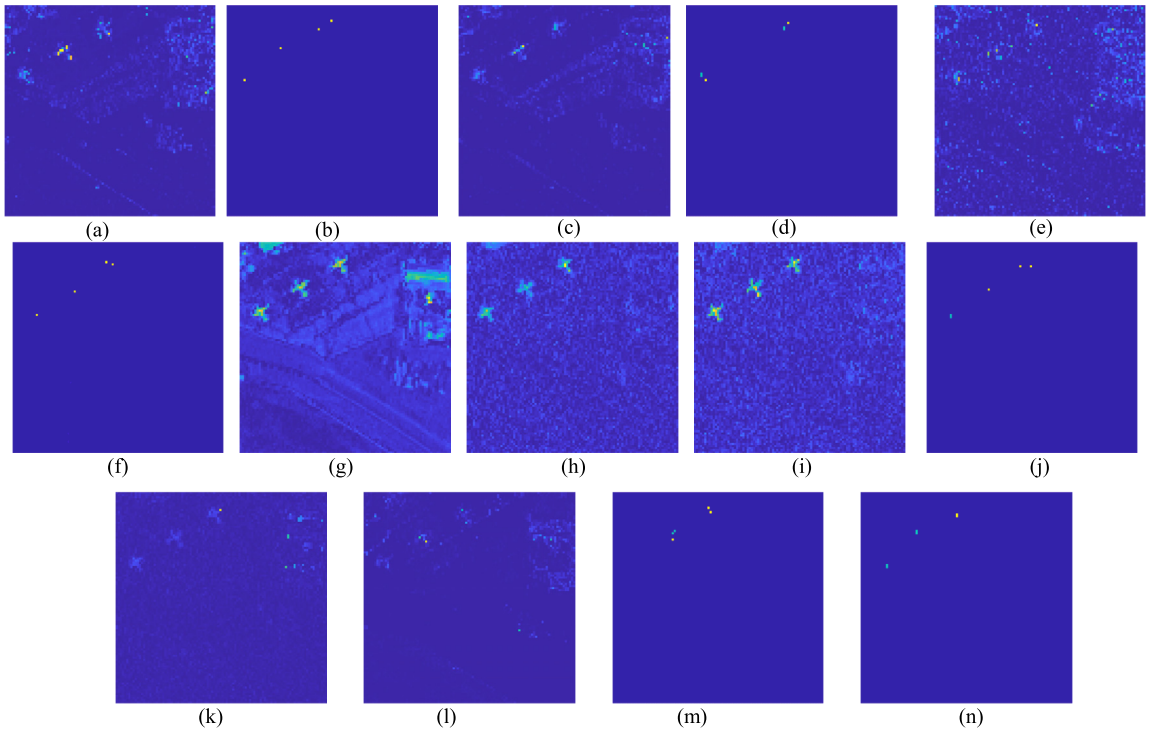


Fig. 10. Detection map of detectors with sampling 5% randomly selected training samples to calculate target signatures. (a) DS-BA-TCIMF. (b) DS-BA-KTCIMF. (c) LRaSMD-BA-TCIMF. (d) LRaSMD BA-KTCIMF. (e) CDASC-BA-TCIMF. (f) CDASC-BA-KTCIMF. (g) OSP. (h) CEM. (i) TCIMF. (j) KTCIMF. (k) SDIA. (l) OSP TD. (m) HCEM. (n) ECEM.

of Fig. 7. Once again, according to Table IV, all kernel-based target detectors, i.e., three versions of BA-KTCIMF, KTCIMF, and HCEM, performed very well, and they were among the three best results. For target detectors without using kernels, ECEM was the best.

VI. TRAINING SAMPLING-BASED TCIMF

The experiments conducted in Sections V-A and V-B assumed that the target signatures are provided by prior knowledge or ground truth. In practical applications, this assumption may not be realistic. However, it is known that OSP, CEM, and KTCIMF are very sensitive to their used target knowledge. Thus, this section investigates this issue using training samples instead of complete ground truth to calculate target signatures. Since the HYDICE panel dataset has only 19 R panel pixels that are too small to be used as a training sample pool, in this case, the San Diego dataset in Fig. 4 was used for this purpose where training samples were randomly selected from the 85 pixels of airplanes to calculate the desired airplane signatures. Two scenarios were considered: 20% random sampling and 5% random sampling from the 85 pixels. Figs. 9 and 10 show the anomaly abundance detection maps of DS-BA-TCIMF, DS-BA-KTCIMF, LRaSMD-BA-TCIMF, LRaSMD-BA-KTCIMF, CDASC-TCIMF, CDASC-KTCIMF, OSP, CEM, ICIMF, KTCIMF, SDIA, OSPTD, HCEM, and ECEM using 20% and 5% randomly selected training samples respectively. It is very interesting to note that except OSP, CEM, and TCIMF that were not affected much by training

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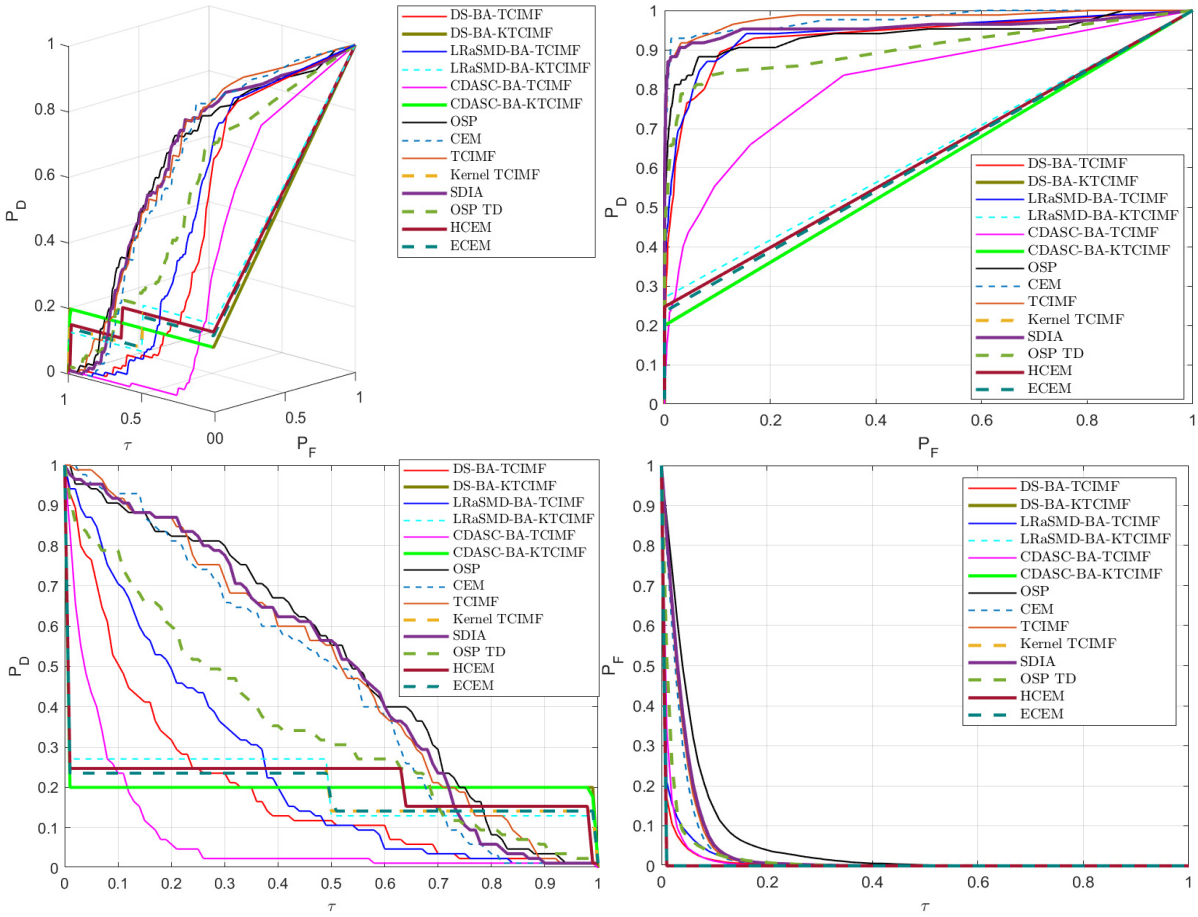


Fig. 11. 3-D ROC curve of detectors. 3-D ROC curves and 2-D ROC curves in the San Diego data scene with a 20% of sampling rate.

TABLE V

AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES IN FIG. 10 FOR SAN DIEGO DETECTION RESULTS

detector	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{SNPR}	AUC_{ODP}	Time(s)
DS-BA-TCIMF	0.8846	0.1096	**0.0111**	0.9942	0.8735	0.0985	9.8163	0.9831	9.0423
DS-BA-KTCIMF	0.6000	0.2030	**0.0050**	0.8030	0.5950	0.1980	**40.6000**	0.7980	5550.7583
LRaSMD-BA-TCIMF	0.9425	0.2501	0.0159	1.1926	**0.9266**	0.2342	15.7171	1.1767	9.7393
LRaSMD-BA-KTCIMF	0.6353	0.2024	**0.0050**	0.8376	0.6303	0.1974	40.4706	0.8326	1465.1743
CDASC-BA-TCIMF	0.9016	0.1949	0.0193	1.0966	0.8824	0.1757	10.1183	1.0773	**1.5532**
CDASC-BA-KTCIMF	0.5997	0.2028	**0.0050**	0.8025	0.5947	0.1978	**40.4994**	0.7975	1436.5258
OSP [5]	0.9391	**0.5092**	0.0583	**1.4483**	0.8808	**0.4509**	8.7327	1.3900	9.0111
CEM [7-8]	**0.9779**	0.4589	0.0340	1.4369	**0.9439**	0.4249	13.4973	1.4029	**1.0337**
TCIMF [10]	**0.9550**	**0.4906**	0.0552	**1.4456**	0.8999	**0.4354**	8.8900	**1.3904**	5.2571
KTCIMF [18]	0.6235	0.1966	**0.0050**	0.8201	0.6184	0.1916	38.9445	0.8151	608.4449
SDIA [12]	**0.9586**	**0.4931**	0.0396	**1.4517**	**0.9191**	**0.4535**	12.4649	**1.4121**	**1.3202**
OSPTD [14]	0.9035	0.3518	0.0200	1.2553	0.8836	0.3319	17.6343	1.2354	301.3799
HCEM [21]	0.6235	0.2144	**0.0050**	0.8379	0.6185	0.2094	**42.8706**	0.8329	2.7533
ECEM [22]	0.6232	0.1910	**0.0051**	0.8142	0.6181	0.1859	37.4077	0.8091	49.4729

samples, all other target detectors that are performed very well in Fig. 7 have been largely impacted by the use of training samples with their detection performance significantly degraded as the number of training samples is decreased. Figs. 11 and 12 also plot the 3-D ROC curves generated from the results in Figs. 9 and 10 along with their corresponding three 2-D ROC curves. Tables V and VI tabulate the AUC values of eight detection measures presented in Section IV-C for quantitative studies and analysis with best in bold, second best in bold red, and third best in bold blue where KTCIMF suffered from exceedingly high computational complexity with very poor performance, while CEM, TCIMF, and OSP remain very effective with very low computational complexity. More interestingly, various versions of BA-TCIMF were still competitive when 20% of training samples were used but lost their compactivity when only 5% of training samples were used. What is worse is that HCEM and ECEM performed very poorly as long as training samples were used instead of the complete ground truth. These experiments were very crucial because they showed that sufficient

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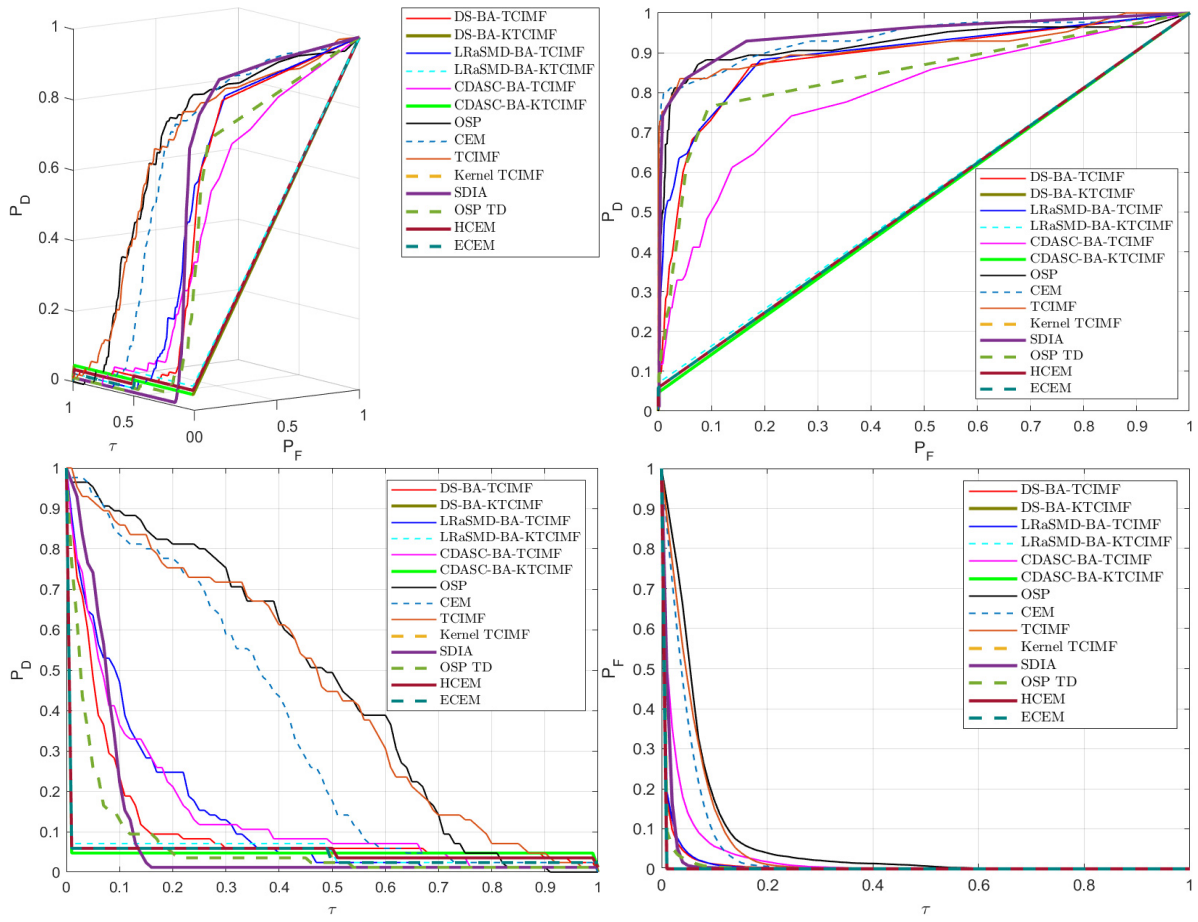


Fig. 12. 3-D ROC curve of detectors. 3-D ROC curves and 2-D ROC curves in the San Diego data scene with a 5% of sampling rate.

TABLE VI

AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES IN FIG. 11 FOR SAN DIEGO DETECTION RESULTS

detector	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{SNPR}	AUC_{ODP}	Time(s)
DS-BA-TCIMF	0.5914	0.1141	0.0507	0.7055	0.5406	0.0634	2.2492	0.6547	8.8871
DS-BA-KTCIMF	0.5235	0.0924	**0.0500**	0.6159	0.4735	0.0424	1.8471	0.5659	5550.7583
LRaSMD-BA-TCIMF	0.7307	0.1482	0.0515	0.8789	0.6792	0.0967	2.8782	0.8274	9.7393
LRaSMD-BA-KTCIMF	0.5353	0.0929	**0.0500**	0.6282	0.4853	0.0429	1.8588	0.5782	1465.1743
CDASC-BA-TCIMF	0.6577	0.1494	0.0580	0.8071	0.5997	0.0914	2.5748	0.7491	**1.5532**
CDASC-BA-KTCIMF	0.5235	0.0929	**0.0500**	0.6165	0.4735	0.0429	1.8588	0.5665	1436.5258
OSP [5]	**0.9232**	**0.4665**	0.0755	**1.3897**	**0.8477**	**0.3910**	**6.1779**	**1.3142**	9.0111
CEM [7-8]	**0.9073**	**0.3453**	0.0591	**1.2526**	**0.8483**	**0.2862**	**5.8448**	**1.1936**	**1.0337**
TCIMF [10]	**0.9096**	**0.4459**	0.0666	**1.3555**	**0.8430**	**0.3793**	**6.6968**	**1.2889**	5.2571
KTCIMF [18]	0.5294	0.0918	0.0500	0.6212	0.4794	0.0418	1.8353	0.5712	608.4449
SDIA [12]	0.6103	0.0824	**0.0505**	0.6927	0.5598	0.0318	1.6303	0.6422	**1.3202**
OSPTD [14]	0.5625	0.0824	**0.0506**	0.6449	0.5119	0.0317	1.6270	0.5942	301.3799
HCEM [21]	0.5294	0.0941	**0.0500**	0.6235	0.4794	0.0441	1.8824	0.5735	2.7533
ECEM [22]	0.5294	0.0865	**0.0500**	0.6158	0.4793	0.0364	1.7280	0.5658	54.1746

ground truth is necessary for HCEM, ECEM, KTCIMF and DS-BA-TCIMF, LRaSMD-TCIMF, and CDASC-TCIMF to perform well, as documented in Tables III and IV. By contrast, when target signatures are calculated by training samples, their performances are degraded drastically and eventually lose their compatibility if the training sample size becomes small.

Three remarks are noteworthy.

1) Despite the fact that KTCIMF performs the best in Figs. 5–8 and Tables III and IV, its slight improvement over BA-TCIMF and extremely high computational complexity prevent it from being applicable to practical applications. This is particularly worse when insufficient training samples are used to calculate the desired target signatures, as demonstrated in Figs. 9–12 and Tables V and VI. Accordingly, the advantages offered by KTCIMF are diminished.
2) TCIMF is a "pixel-based" target detection technique and does not use windows or machine learning/DL techniques to learn BKG. Thus, its applicability is not limited by dataset size. Thus, its effective performance

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has little to do with increasing datasets but rather BKG. It is exactly the key idea of BA-TCIMF that introduces two types of BKG annihilation.

3) Last but not least, DL-based target detection algorithms are not applicable to comparing BA-TCIMF is because BA-TCIMF does not require training samples, while DL does. Comparing BA-TCIMF to DL-based target detection is like comparing apples to oranges. In particular, when DL-based methods are used, they generally use training samples to learn the BKG. By contrast, BA-TCIMF does not need to do so. Instead, it makes use of BA to accomplish BS. This is generally more effective than using DL-based methods to estimate BKG for removal. Specifically, the great advantage of BA-TICMF over the DL-based methods is that BA-TCIMF makes use of DS and unsupervised MX-SVD to find unwanted BKG signatures through the GoDec model. This has been shown by the experiments conducted for the San Diego airport scene in [38, Sect. VIII] where hCEM and ECEM were shown in [20] to outperform machine learning-based methods in BS, such as PTA (prior-based tensor approximation) in [39], LSDM-MoG (low-rank and sparse decomposition with mixture of Gaussians) in [40,] and LRR in [41].

VII. Conclusion

TCIMF is an effective subpixel target detector in many applications. However, its utilization is limited to the requirement of prior knowledge of the undesired target signatures in $\mathbf{U}$. This article presents a generalized TCIMF, called BA-TCIMF that can be implemented in a two-stage process with the first stage annihilating BKG followed by a second stage TDE and improving target BKG suppression (TBS). Depending upon how to annihilate BKG in the first stage, three versions of BA-TCIMF, DS-BA-TCIMF, LRaSMD-BA-TCIMF, and CDA-BA-TCIMF, are derived in three different ways. The main contributions of this article can be described as follows.

1) It develops BA-TCIMF as a two-stage target detector that includes TCIMF, SDIA, and OSP-TD as special cases. Specifically, three versions of BA-TCIMF are derived, that is, DS-BA-TCIMF, LRaSMD-BA-TCIMF, and CDA-TCIMF.
2) In the first stage, BA-TCIMF annihilates BKG prior to target detection. That is, DS-BA-TCIMF takes care of BKG by DS; LRaSMD-BA-TCIMF removes BKG by the GoDec-produced low-rank space $\mathbf{L}$ via $P_{\mathbf{L}}^{\perp}$; and CDA-BA-TCIMF nulls out the CDA-generated PC space as BKG space.
3) In the second stage, BA-TCIMF extracts additional unwanted BKG signatures $\mathbf{B}_m$ to augment the undesired target signature matrix $\mathbf{U}$ to $\mathbf{U}\mathbf{B}_m$ to improve TD.
4) In the meantime, at the second stage, BA-TCIMF further suppresses target BKG by inverting the correlation matrix of BA data, $\mathbf{R}_{\mathrm{BA}}$. DS-BA-TCIMF operates $\mathbf{R}_{\mathrm{BA}}$ in the sphered joint $\mathbf{L}+\mathbf{S}$ space, $\hat{\mathbf{L}}+\hat{\mathbf{S}}$, while LRaSMD-BA-TCIMF operates $\mathbf{R}_{\mathrm{BA}}$ in the joint $\mathbf{L}+\mathbf{S}$ space. Different from DS-BA-TCIMF and LRaSMD-BA-TCIMF, CDA-BA-TCIMF operates $\mathbf{R}_{\mathrm{BA}}$ in the joint $\mathbf{PC}+\mathbf{IC}$ component space.
5) Despite that KTCIMF can be also used to extend to BA-KTCIMF to improve the performance of BA-TCIMF, such effort is not worth. This is because BA-TCIMF has already been demonstrated to be very effective, and BA-KTCIMF may only improve its performance slightly but not much. Most importantly, the computational complexity of using kernels is exceedingly expensive, specifically, kernelizing the correlation matrix $\mathbf{R}_{\mathrm{BA}}$.

As a final comment, many researchers may think that target detection and anomaly detection are similar detection tasks and expect that one can derive the other. As a matter of fact, this is certainly not true. More specifically, target detection is active detection with known target knowledge as opposed to anomaly detection, which is passive detection with no prior target knowledge. There are significant differences between target detection and anomaly detection, as detailed in [20], [42], and [23]. Consequently, one cannot simply apply target detection to another detection and vice versa.

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